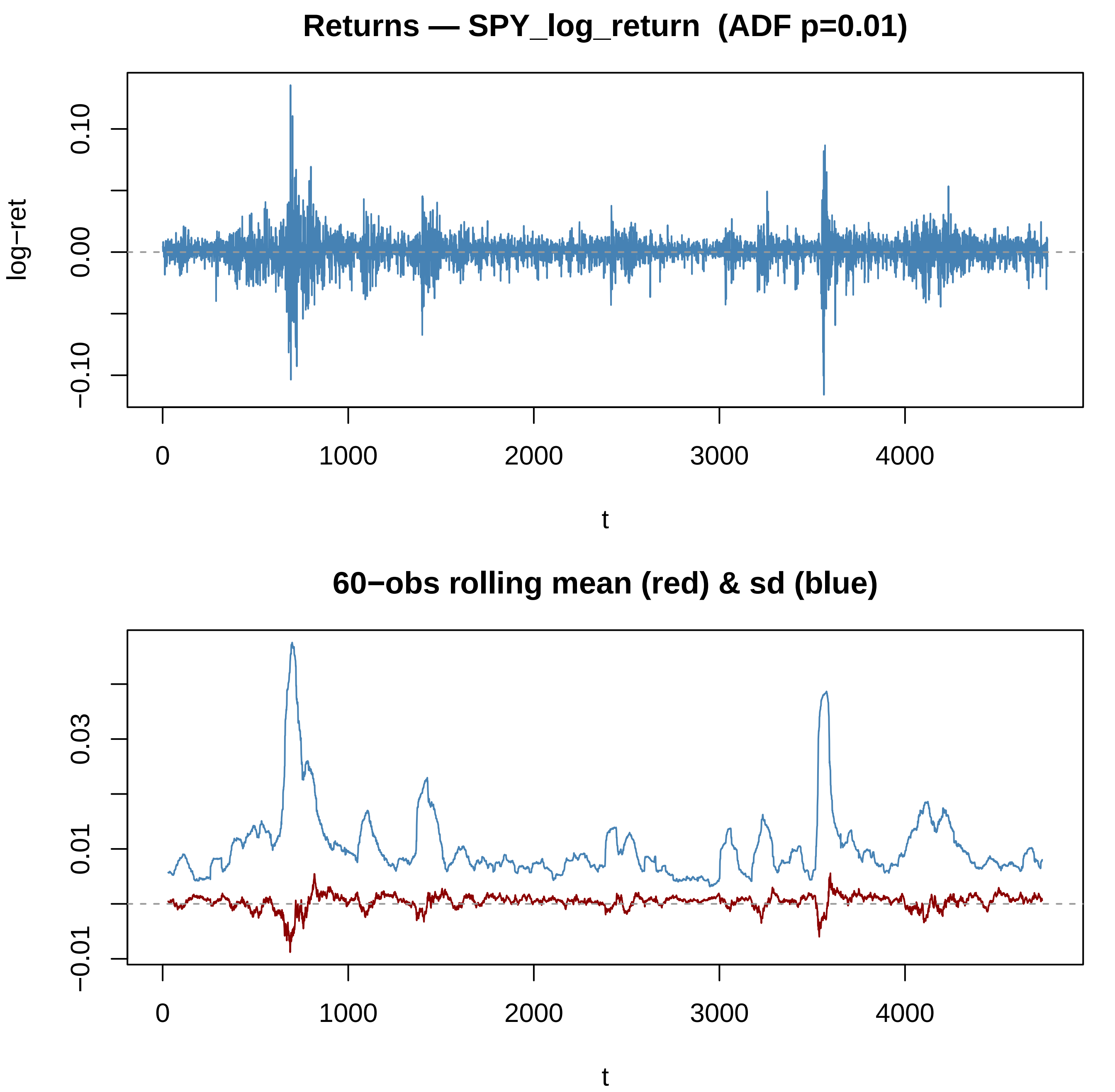
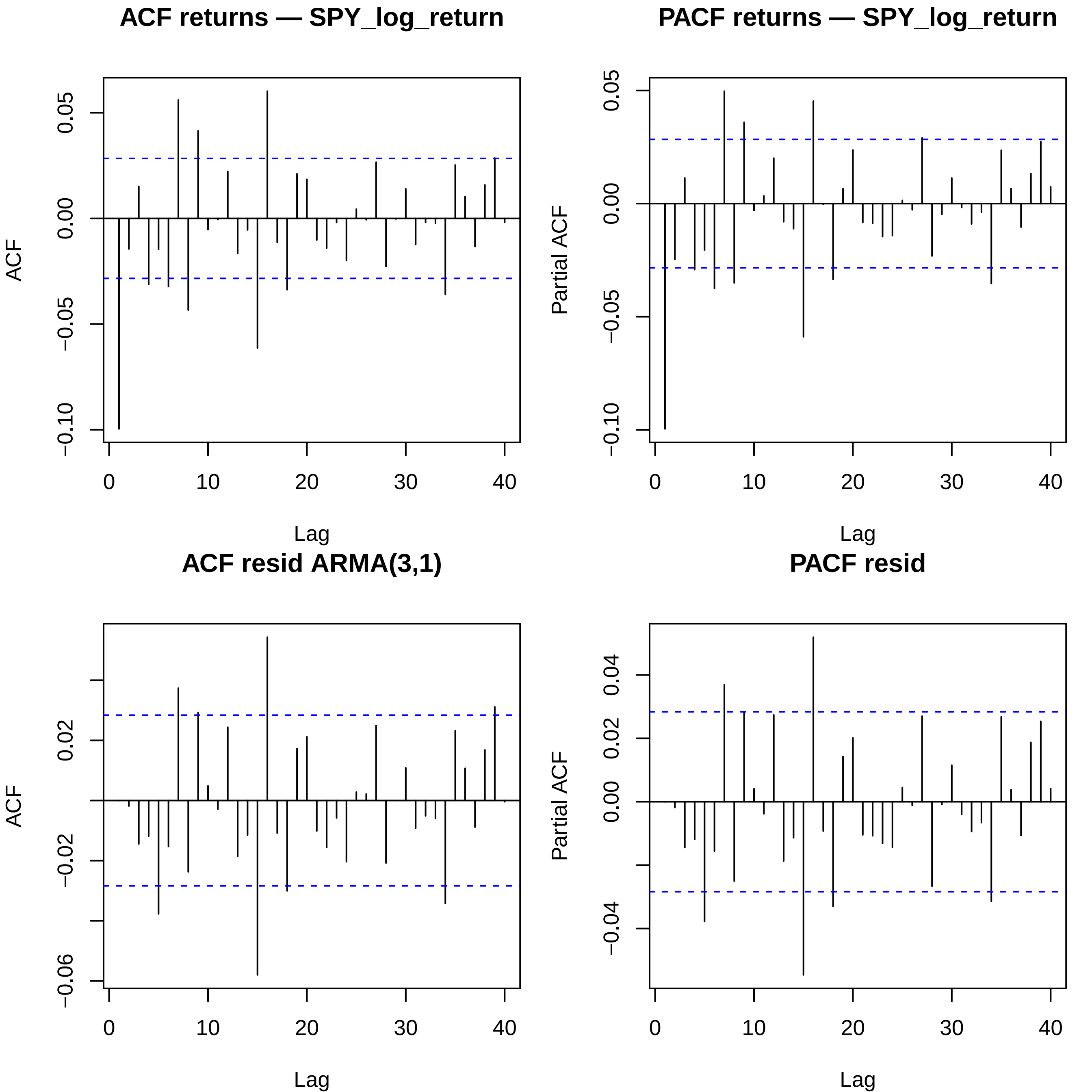
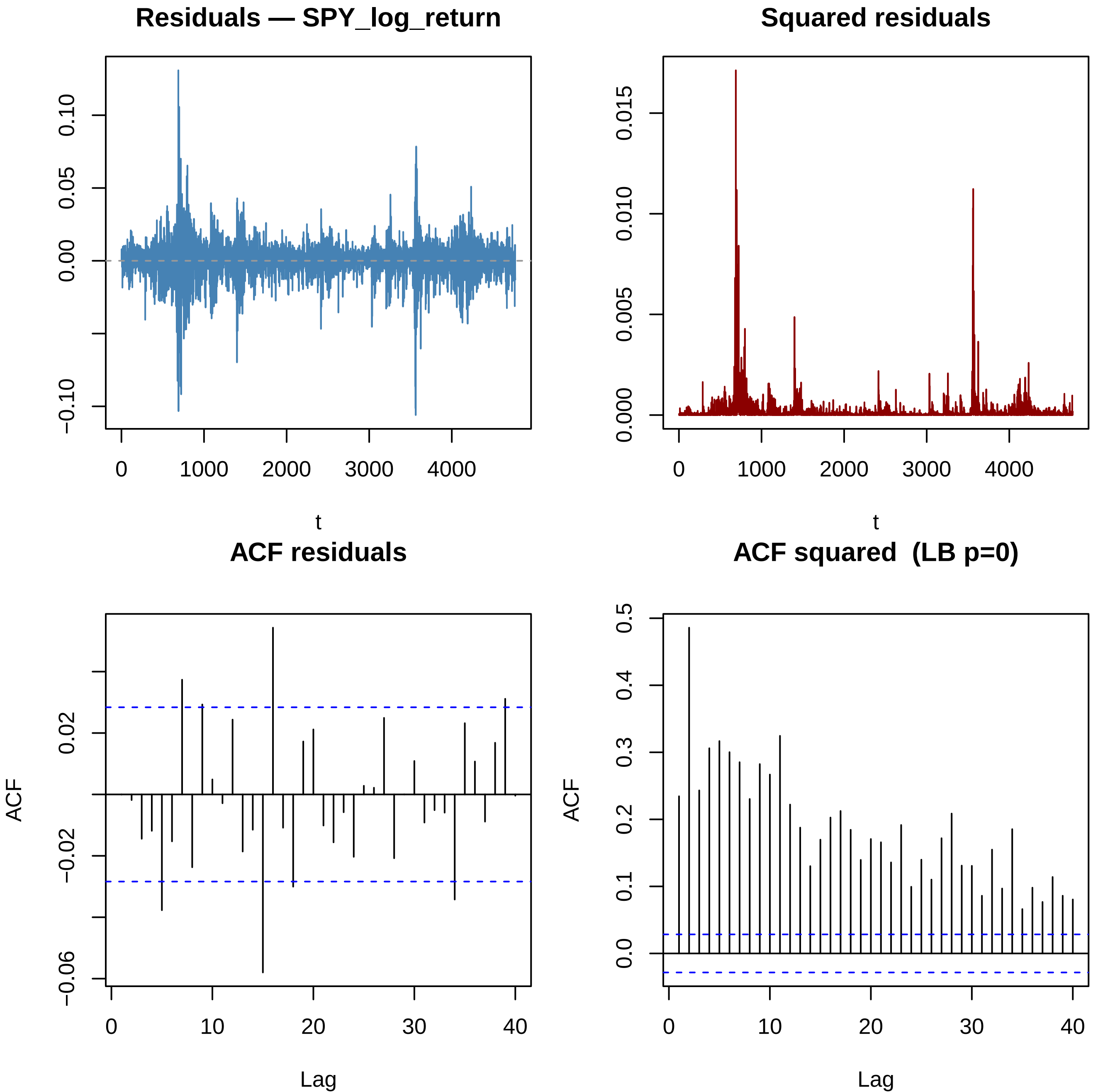


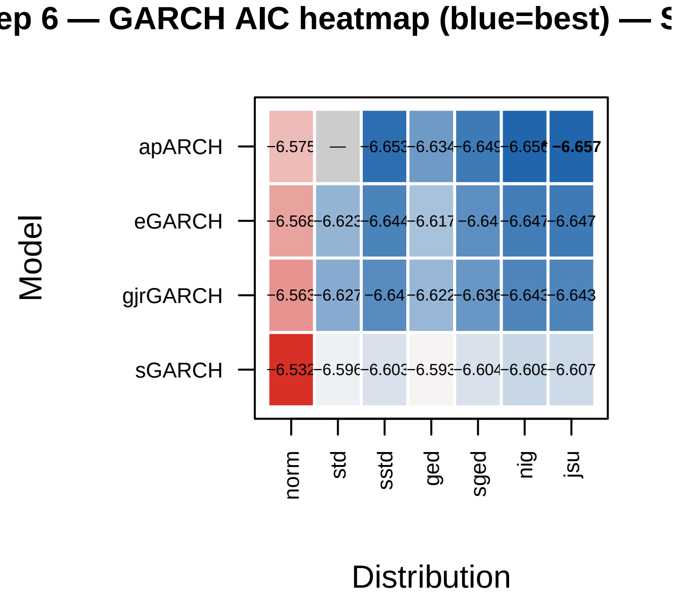
Returns — SPY_log_return (ADF p=0.01)
log−ret
0.10
0.00
−0.10
0
1000
2000
3000
4000
t
60−obs rolling mean (red) & sd (blue)
0.03
0.01
−0.01
0
1000
2000
3000
4000
t



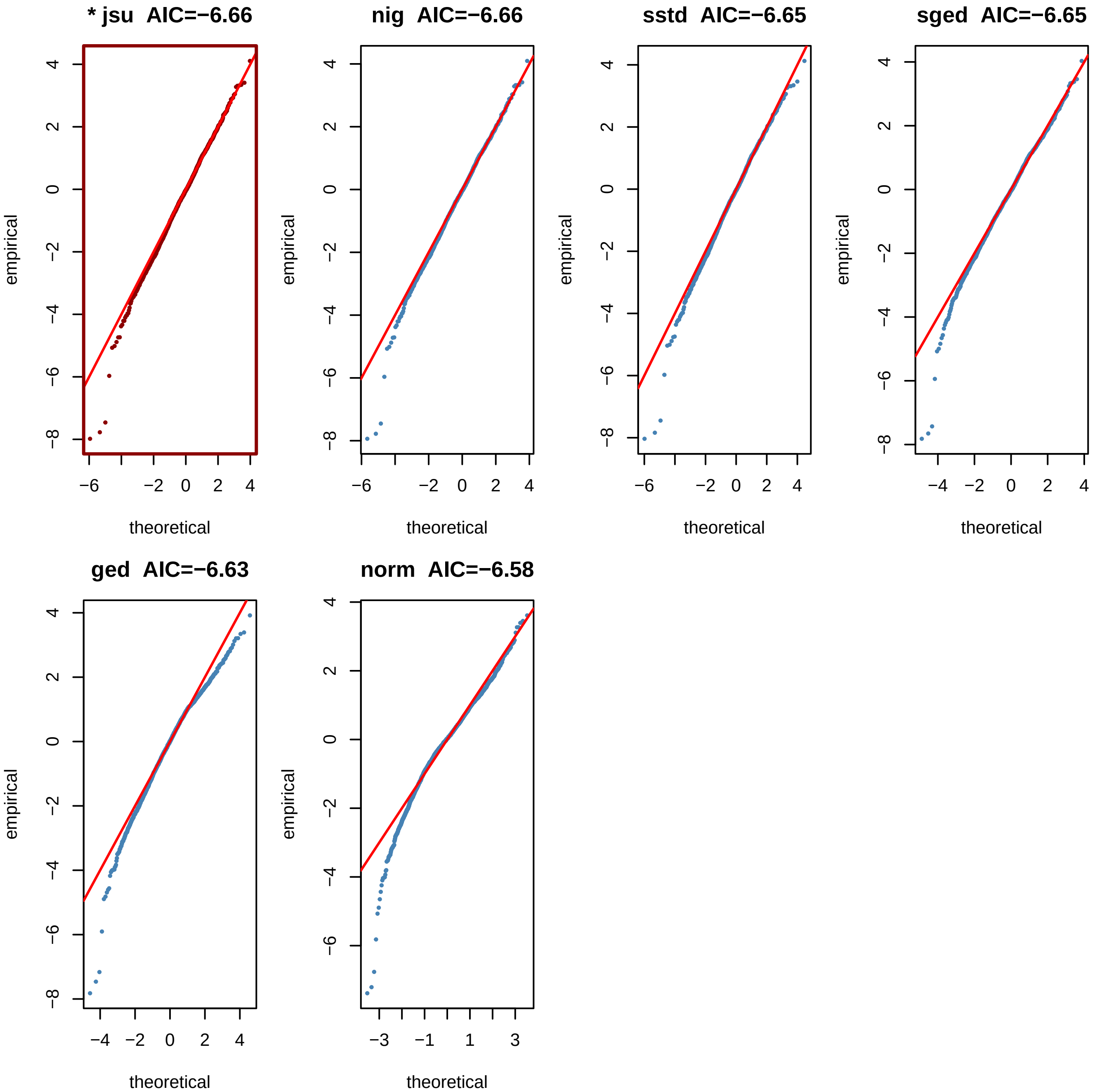
ACF returns — SPY_log_return
PACF returns — SPY_log_return
ACF resid ARMA(3,1)
PACF resid
ACF
Partial ACF
Lag
0.05
0.00
−0.05
−0.10
0.04
0.02
−0.02
−0.04
−0.06
0
10
20
30
40



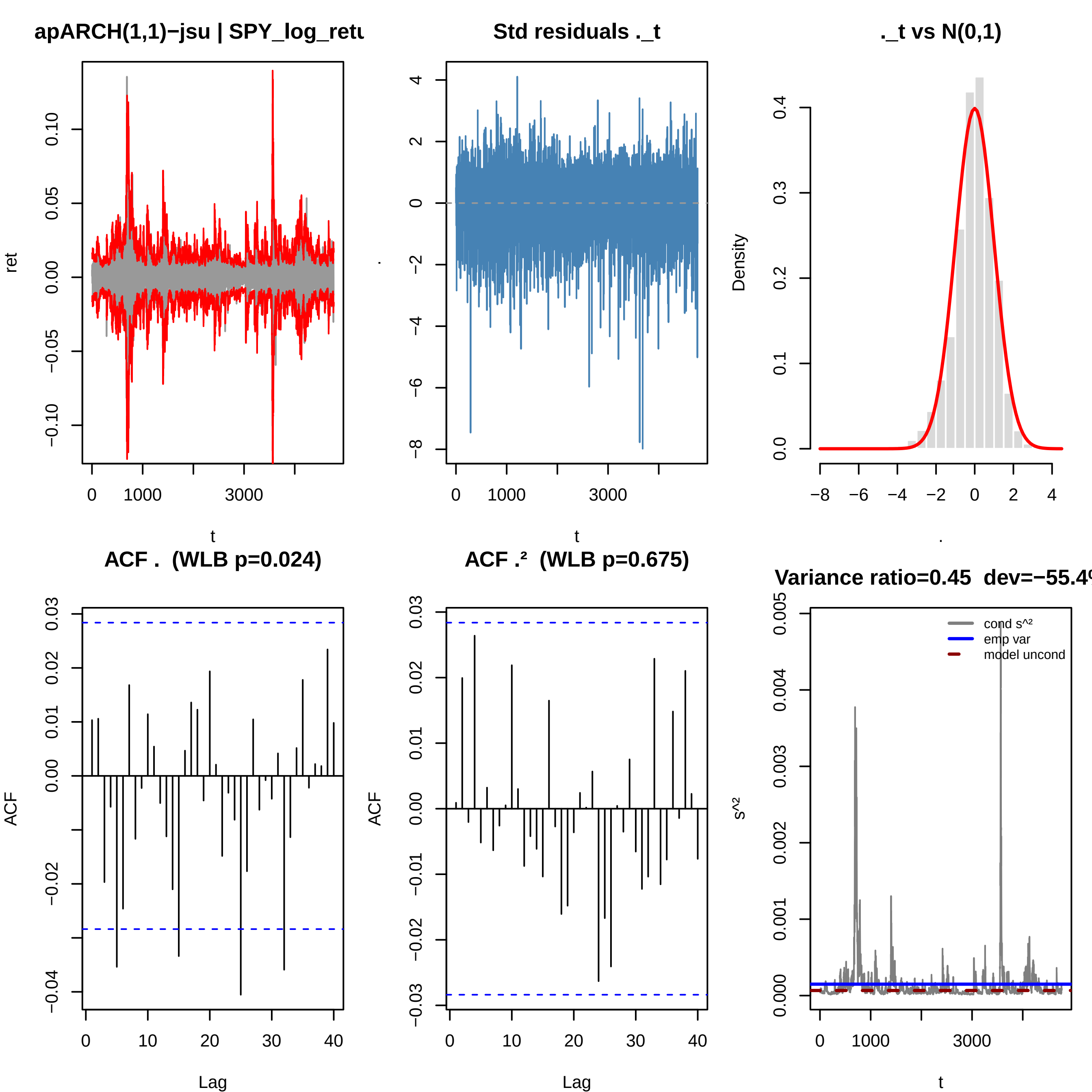
Residuals — SPY_log_return
−0.10
0.00
0.05
0.10
0
1000
2000
3000
4000
t
Squared residuals
0.000
0.005
0.010
0.015
0
1000
2000
3000
4000
t
ACF residuals
ACF
−0.06
−0.02
0.02
0
10
20
30
40
Lag
ACF squared (LB p=0)
ACF
0.0
0.1
0.2
0.3
0.4
0.5
0
10
20
30
40
Lag



ep 6 — GARCH AIC heatmap (blue=best) — S
Model
apARCH
eGARCH
gjrGARCH
sGARCH
−6.575
—
−6.653
−6.634
−6.649
−6.656
−6.657
−6.568
−6.623
−6.644
−6.617
−6.64
−6.647
−6.647
−6.563
−6.627
−6.64
−6.622
−6.636
−6.643
−6.643
−6.532
−6.596
−6.603
−6.593
−6.604
−6.608
−6.607
norm
std
sstd
ged
sged
nig
jsu
Distribution

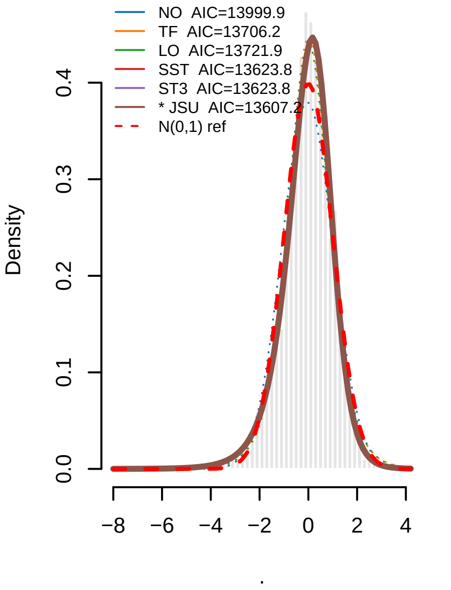


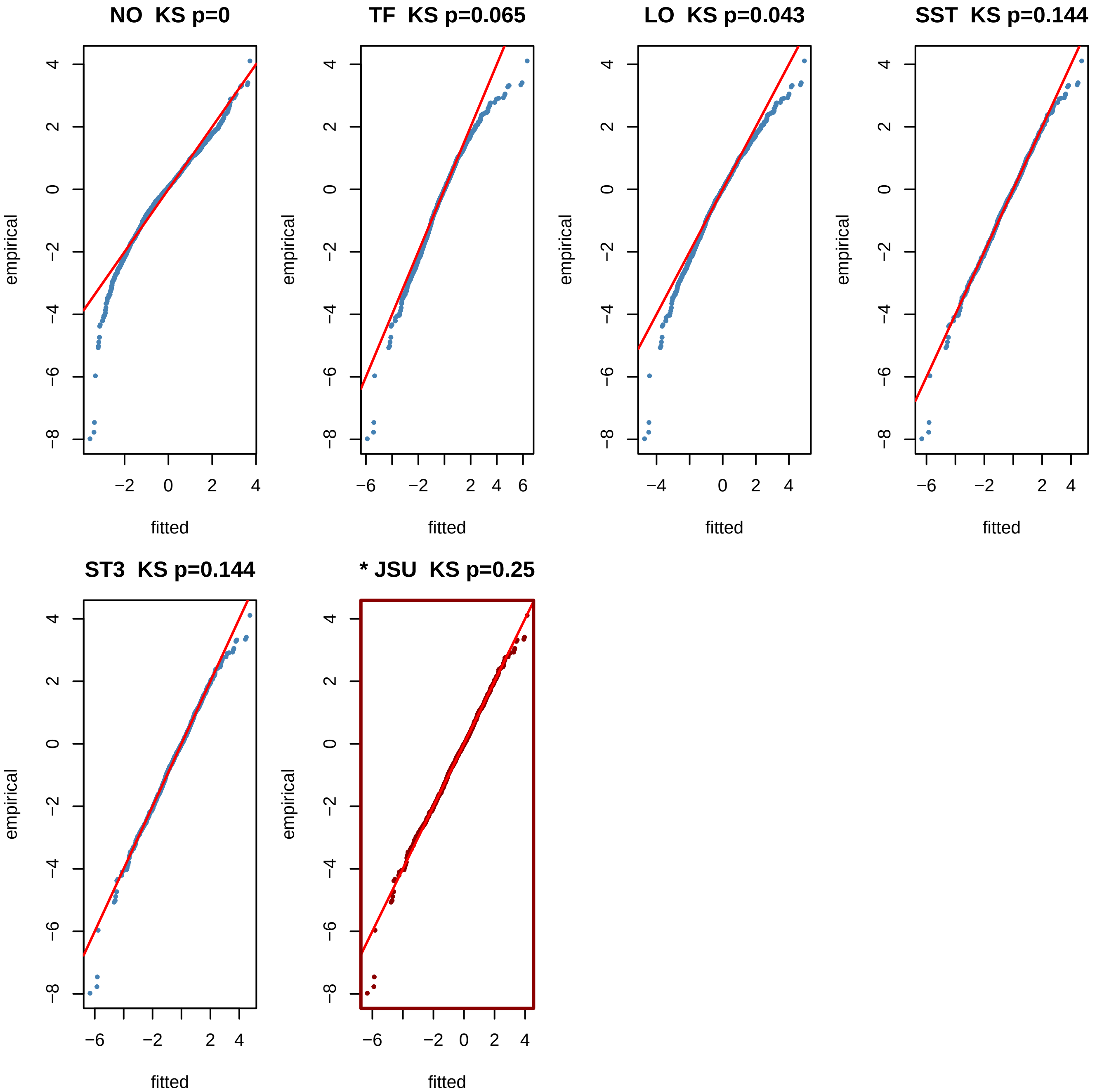
* jsu AIC=−6.66
nig AIC=−6.66
sstd AIC=−6.65
sged AIC=−6.65
ged AIC=−6.63
norm AIC=−6.58
empirical
theoretical

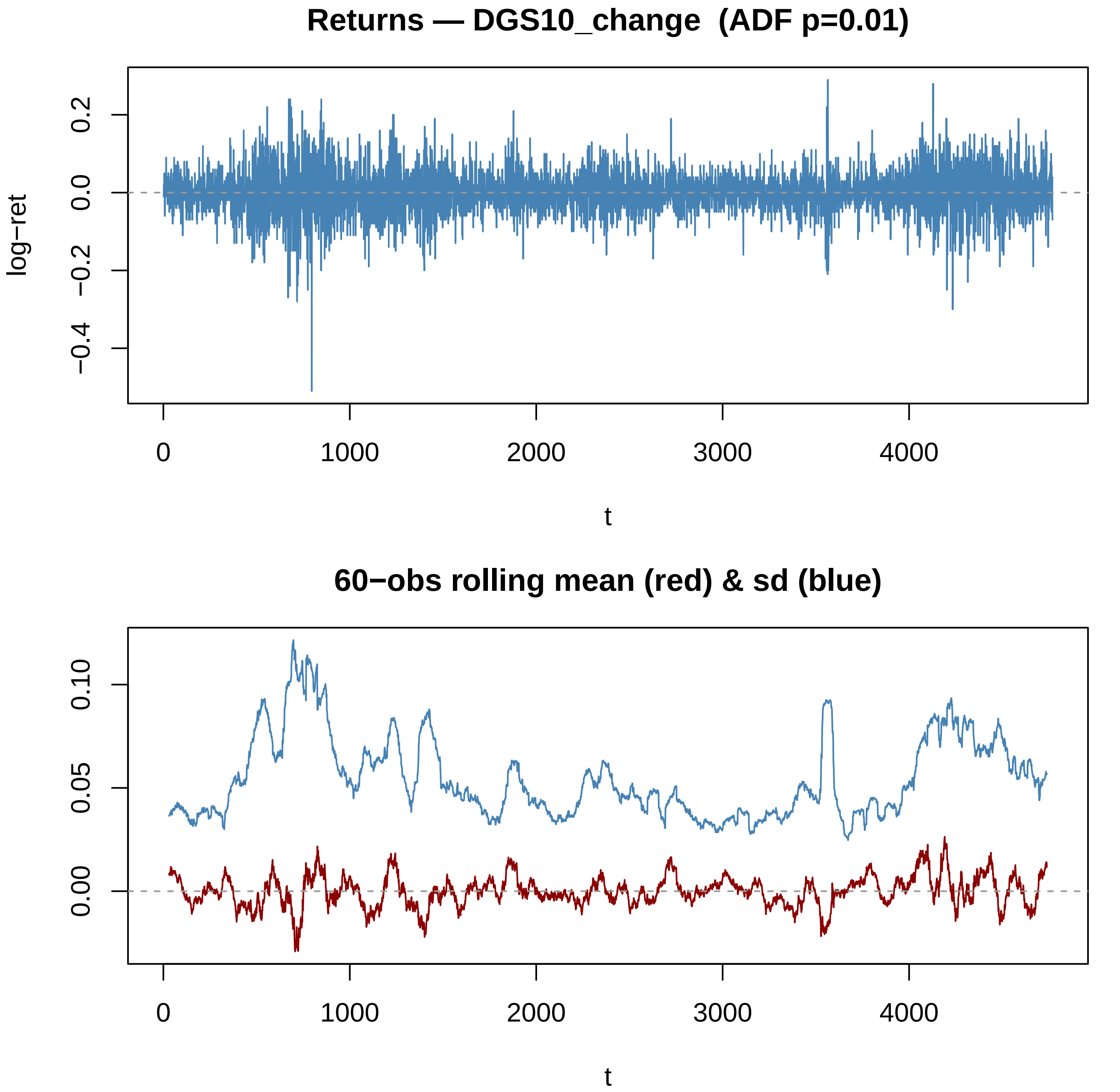


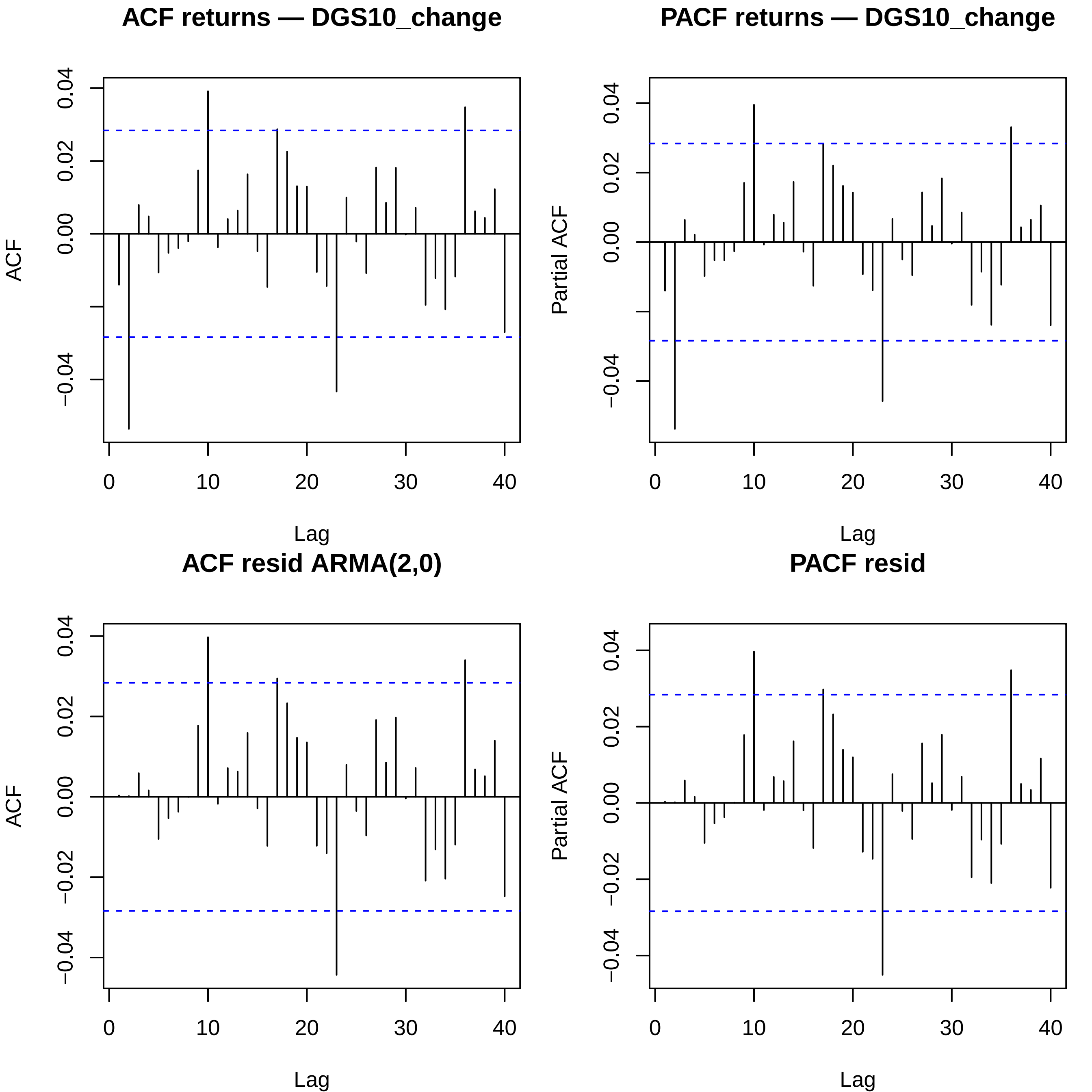
apARCH(1,1)−jsu | SPY_log_retu
Std residuals ._t
._t vs N(0,1)
ACF . (WLB p=0.024)
ACF .² (WLB p=0.675)
Variance ratio=0.45 dev=−55.4
ret
t
Density
ACF
Lag
s^2
cond s^2
emp var
model uncond

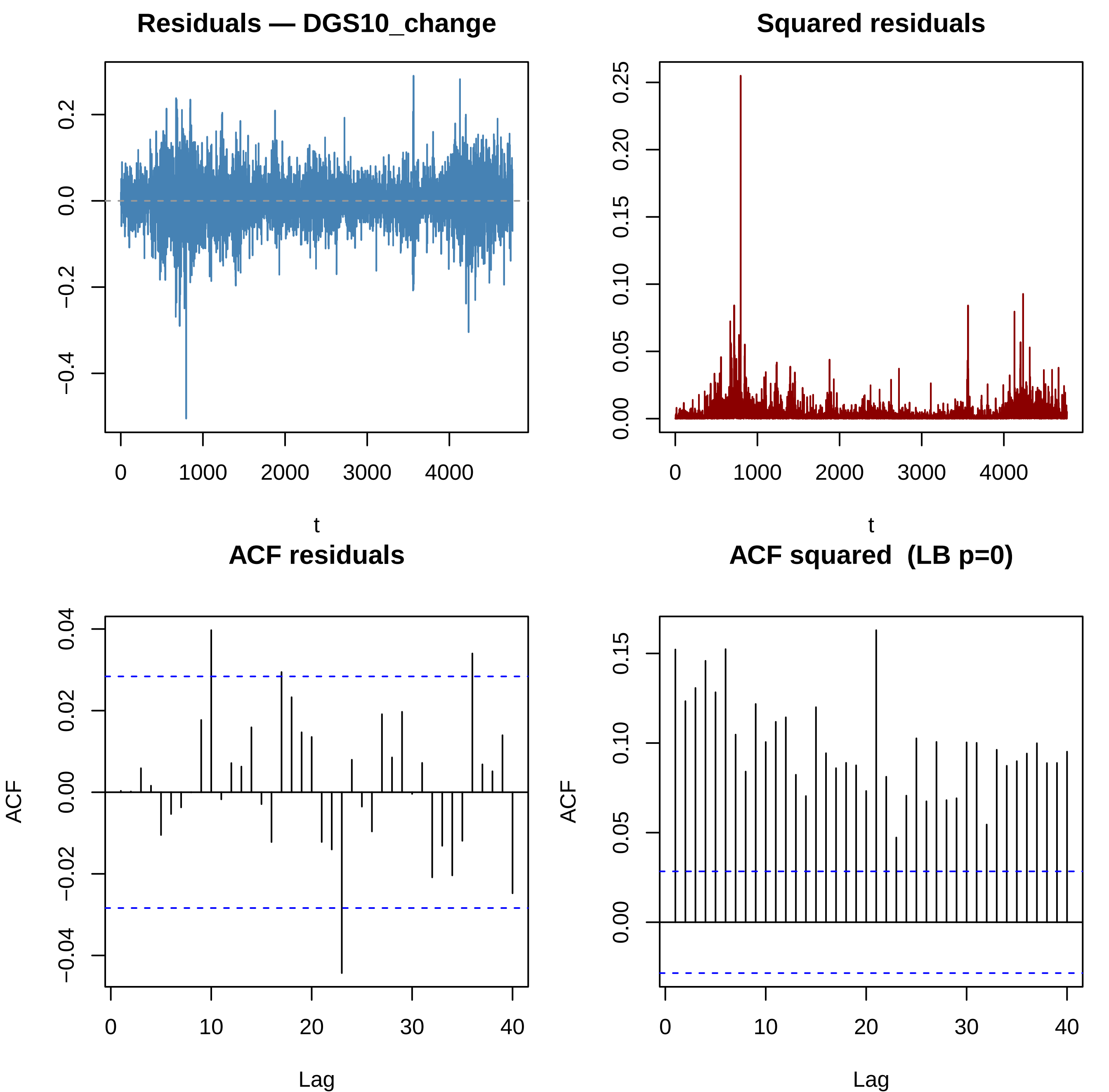
Step 8 — PIT density fits to ._t — SPY_l



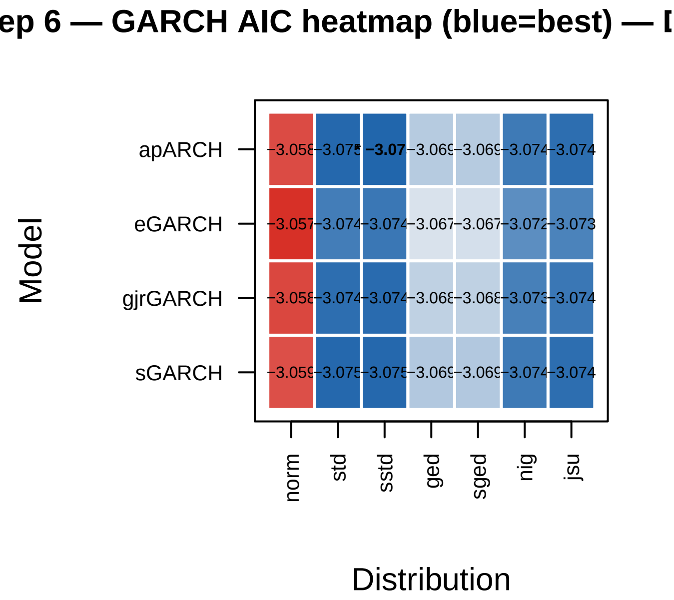

NO KS p=0
TF KS p=0.065
LO KS p=0.043
SST KS p=0.144
ST3 KS p=0.144
* JSU KS p=0.25
empirical
fitted


Returns — DGS10_change (ADF p=0.01)
log−ret
0.2
0.0
−0.2
−0.4
0
1000
2000
3000
4000
t
60−obs rolling mean (red) & sd (blue)
0.10
0.05
0.00
0
1000
2000
3000
4000
t

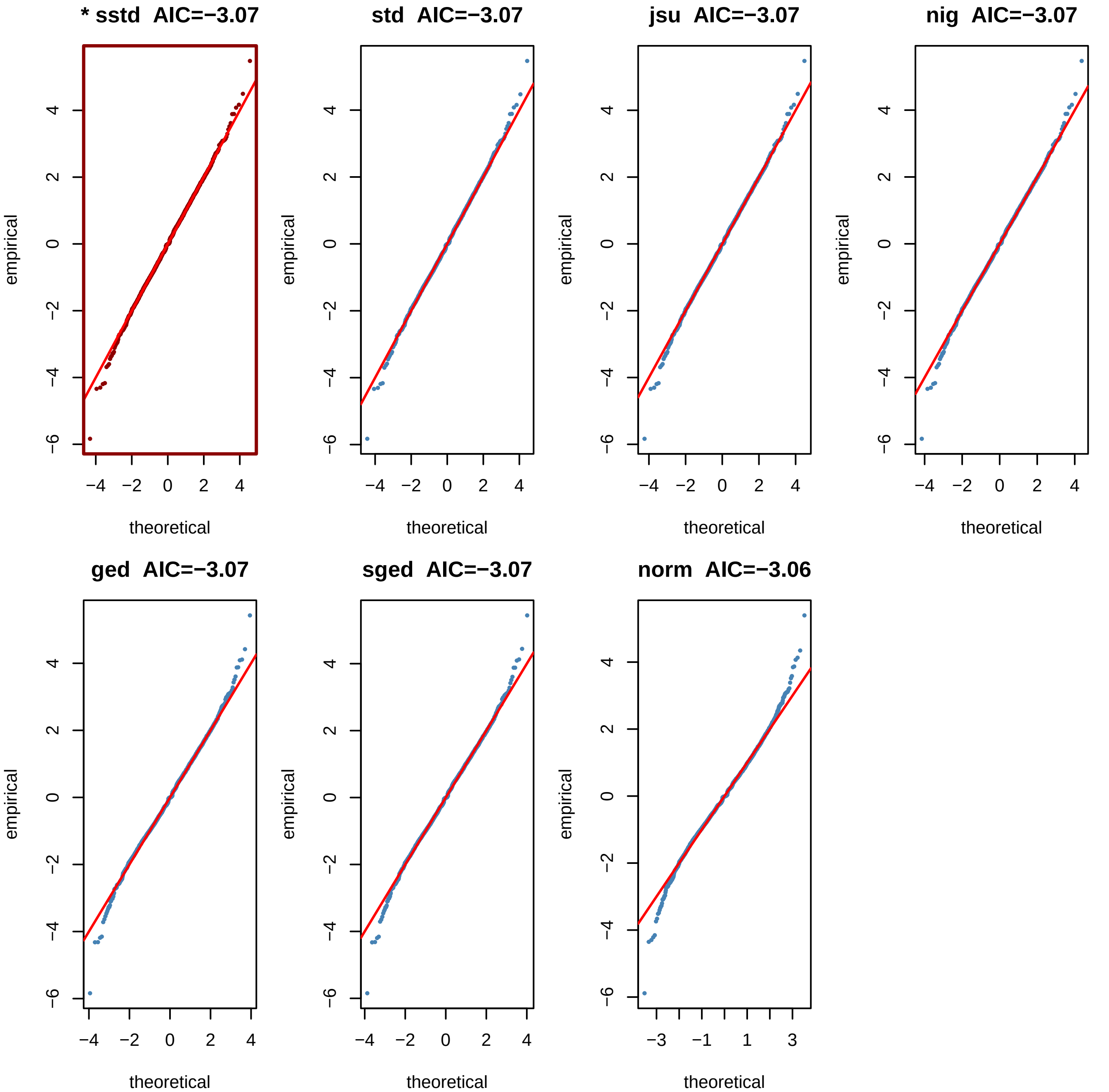

ACF returns — DGS10_change
PACF returns — DGS10_change
ACF resid ARMA(2,0)
PACF resid
ACF
Partial ACF
Lag
0
10
20
30
40
0.04
0.02
0.00
−0.02
−0.04



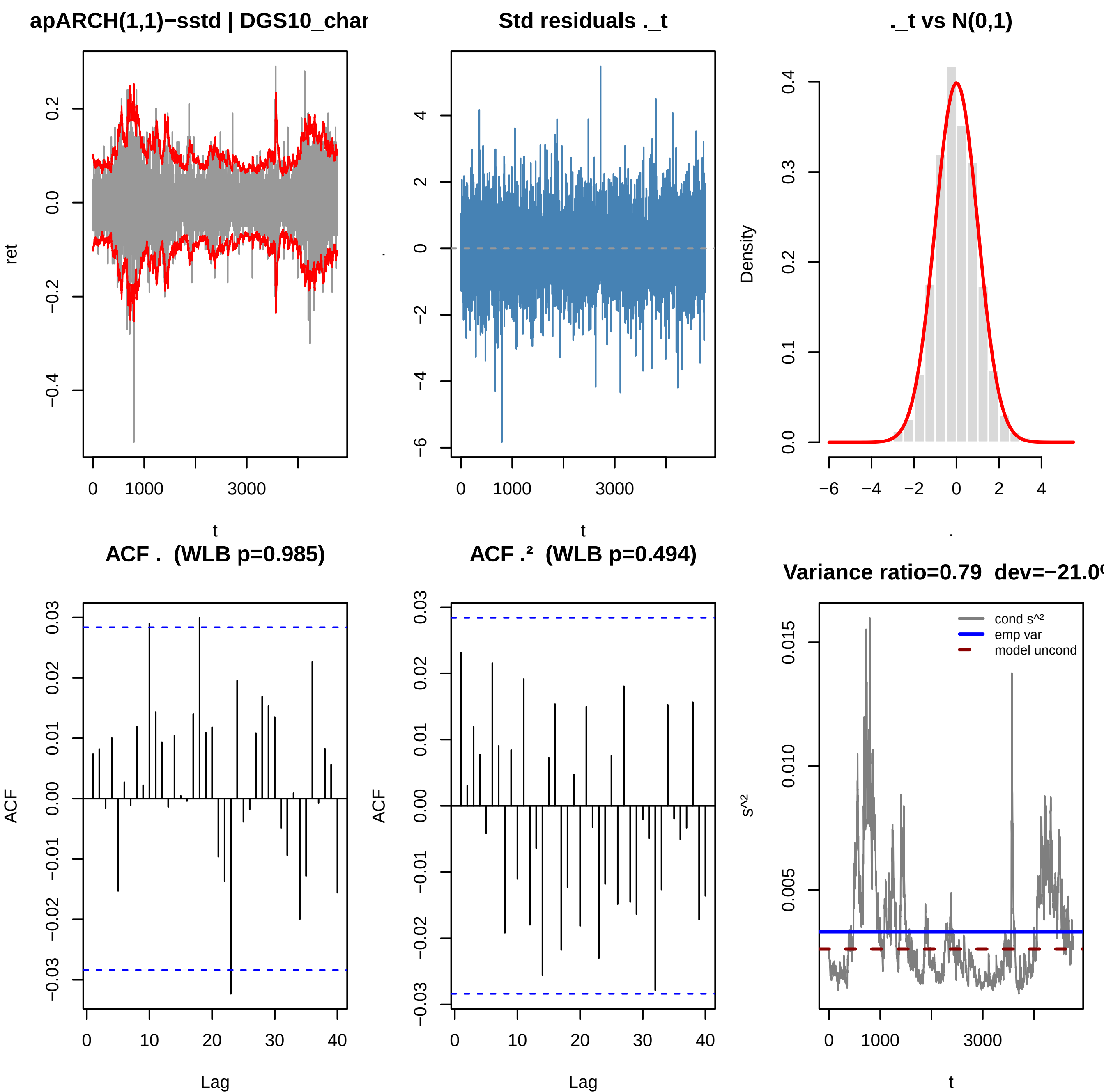
Residuals — DGS10_change
Squared residuals
ACF residuals
ACF squared (LB p=0)
t
Lag
ACF



ep 6 — GARCH AIC heatmap (blue=best) — D
Model
apARCH
eGARCH
gjrGARCH
sGARCH
−3.058 −3.075 −3.07 −3.069 −3.069 −3.074 −3.074
−3.057 −3.074 −3.074 −3.067 −3.067 −3.072 −3.073
−3.058 −3.074 −3.074 −3.068 −3.068 −3.073 −3.074
−3.059 −3.075 −3.075 −3.069 −3.069 −3.074 −3.074
norm
std
sstd
ged
sged
nig
jsu
Distribution

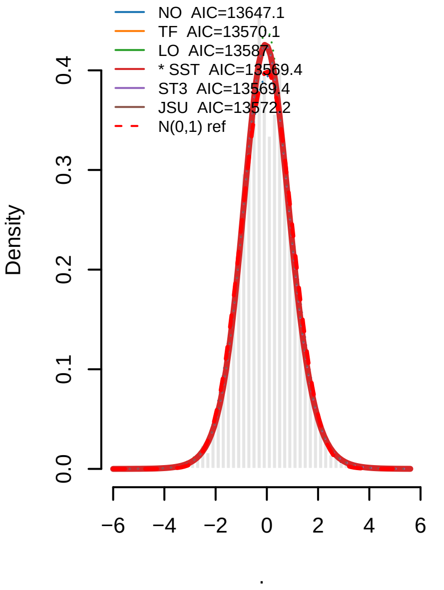


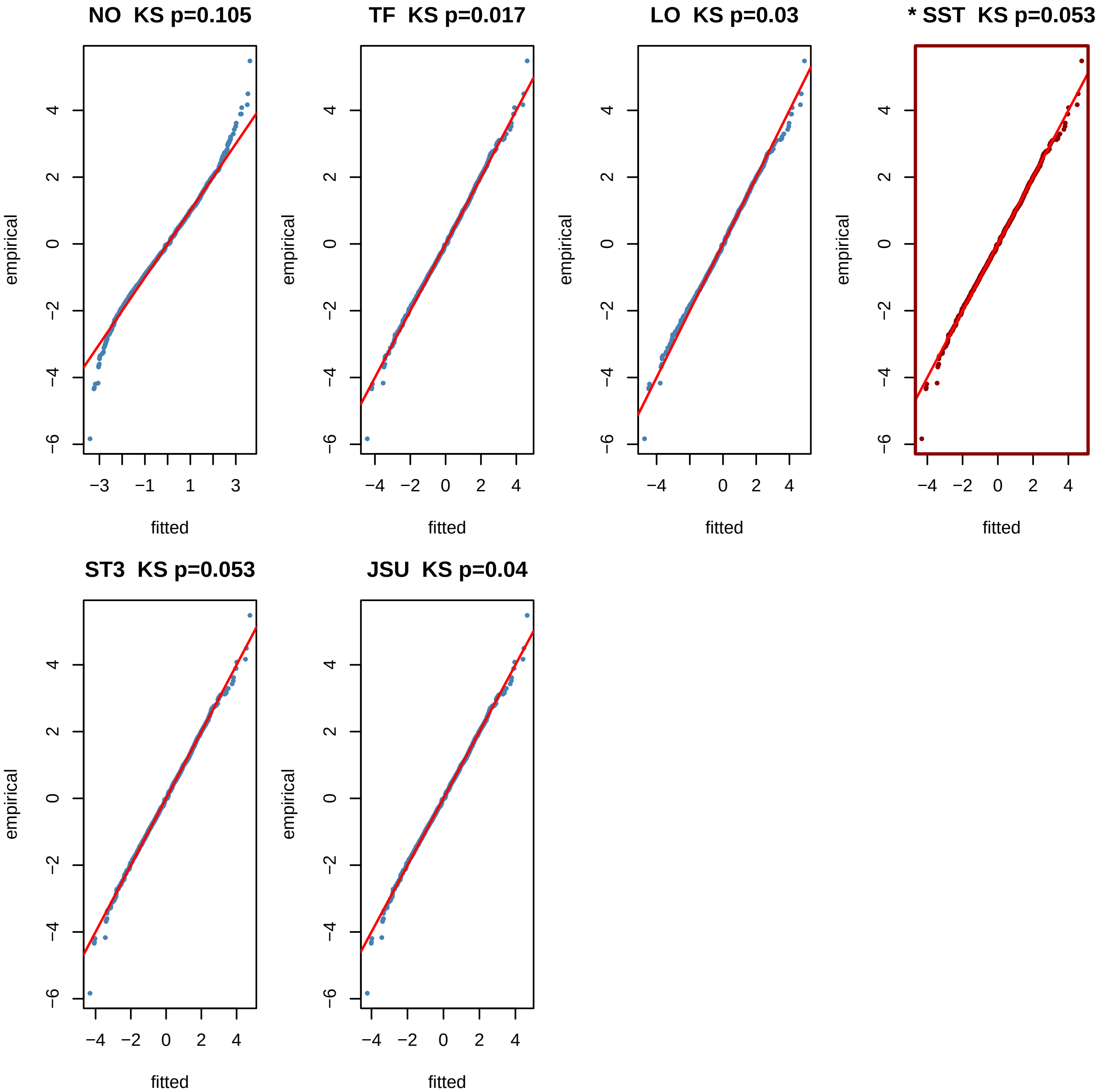
* sstd AIC=−3.07
std AIC=−3.07
jsu AIC=−3.07
nig AIC=−3.07
ged AIC=−3.07
sged AIC=−3.07
norm AIC=−3.06
empirical
theoretical

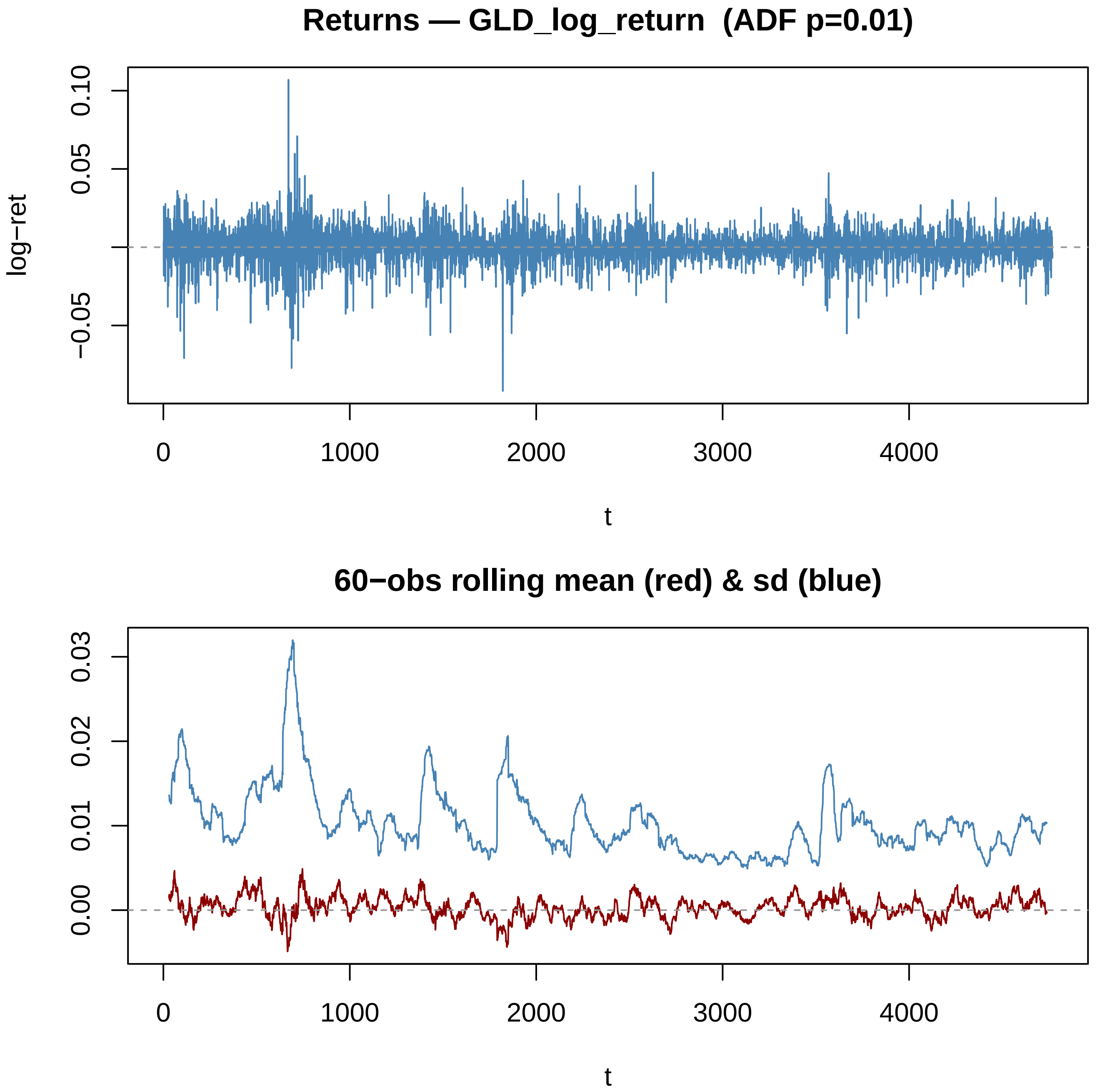


apARCH(1,1)−sstd | DGS10_char
ret
t
Std residuals ._t
._t vs N(0,1)
Density
ACF . (WLB p=0.985)
ACF .² (WLB p=0.494)
Variance ratio=0.79 dev=−21.0
cond s^2
emp var
model uncond
ACF
Lag
s^2

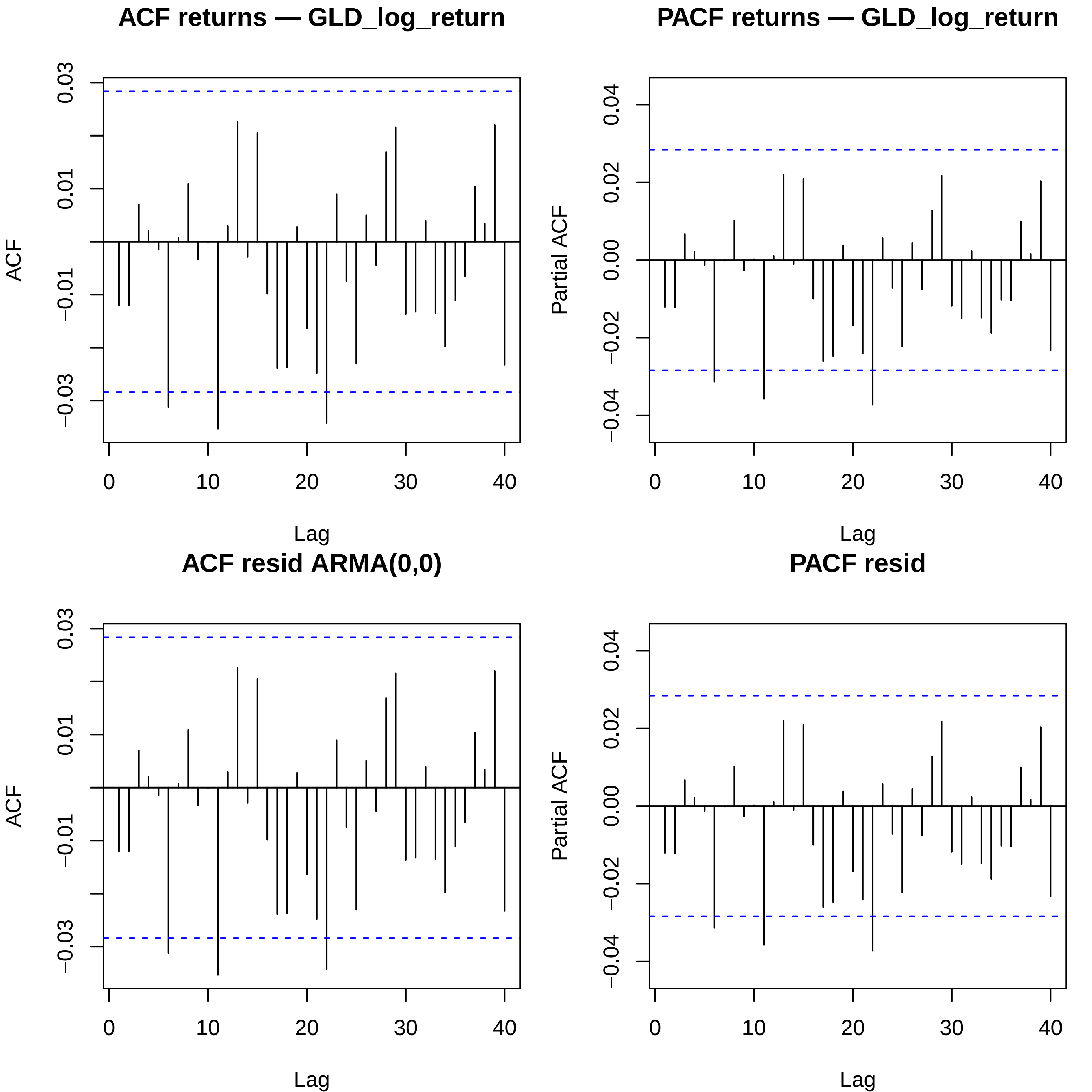
p 8 — PIT density fits to ._t — DGS1



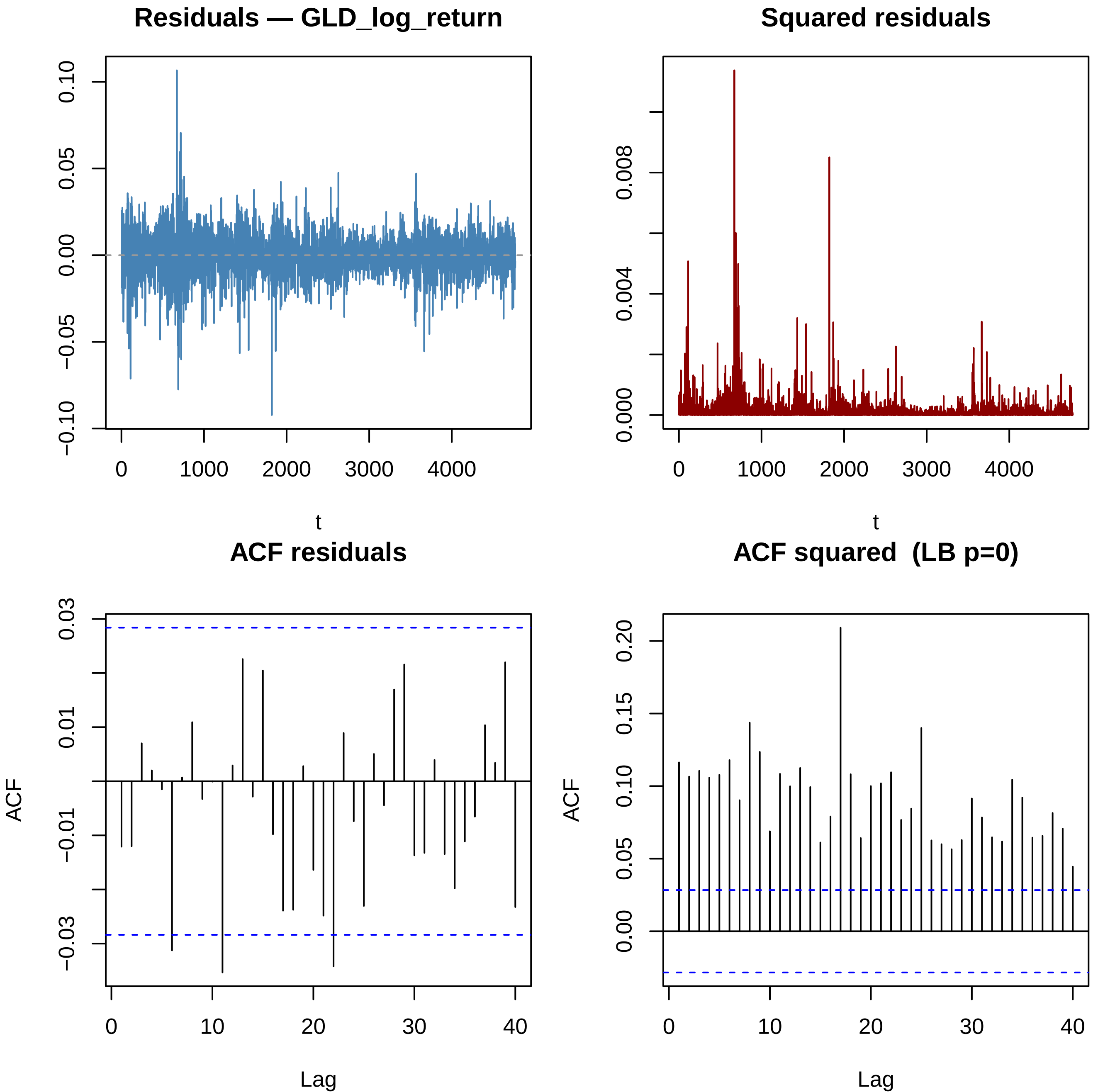

NO KS p=0.105
TF KS p=0.017
LO KS p=0.03
* SST KS p=0.053
ST3 KS p=0.053
JSU KS p=0.04
empirical
fitted
−6
−4
−2
0
2
4
−3
−1
1
3



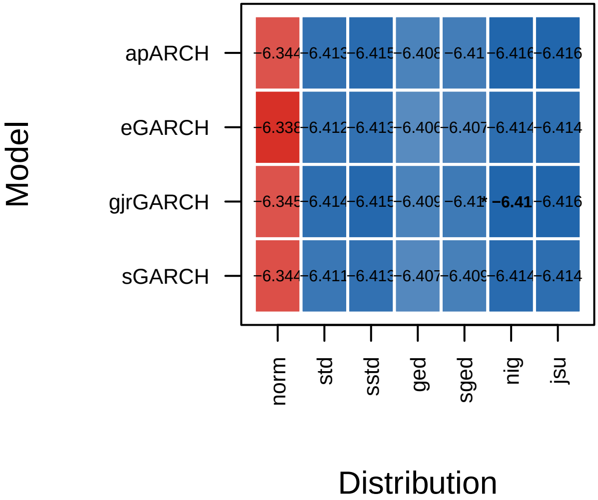
Returns — GLD_log_return (ADF p=0.01)
log−ret
0.10
0.05
−0.05
0
1000
2000
3000
4000
t
60−obs rolling mean (red) & sd (blue)
0.03
0.02
0.01
0.00
0
1000
2000
3000
4000
t

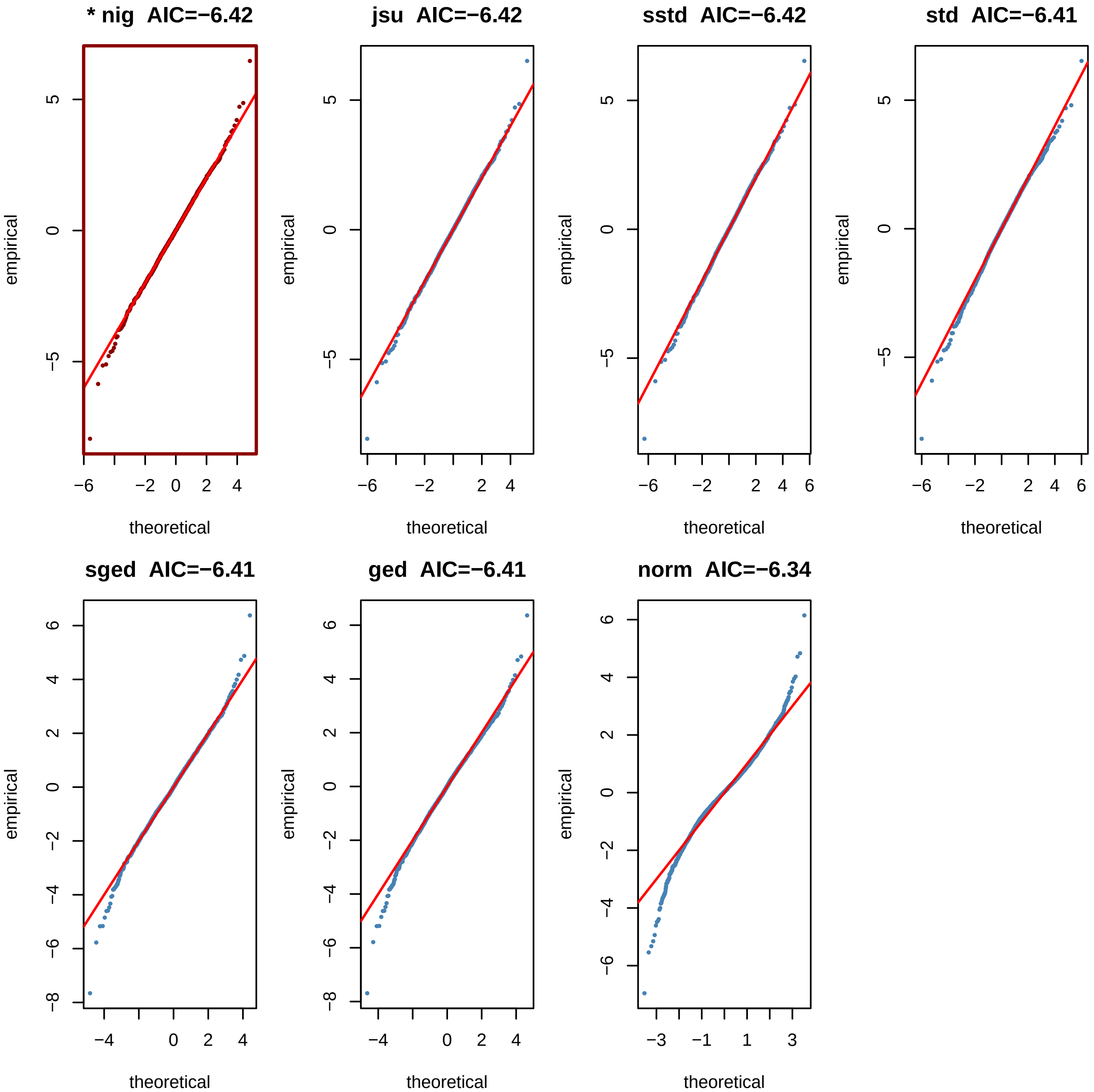


ACF returns — GLD_log_return
PACF returns — GLD_log_return
ACF resid ARMA(0,0)
PACF resid
ACF
Partial ACF
Lag
0.03
0.01
−0.01
−0.03
0.04
0.02
0.00
−0.02
−0.04
0
10
20
30
40

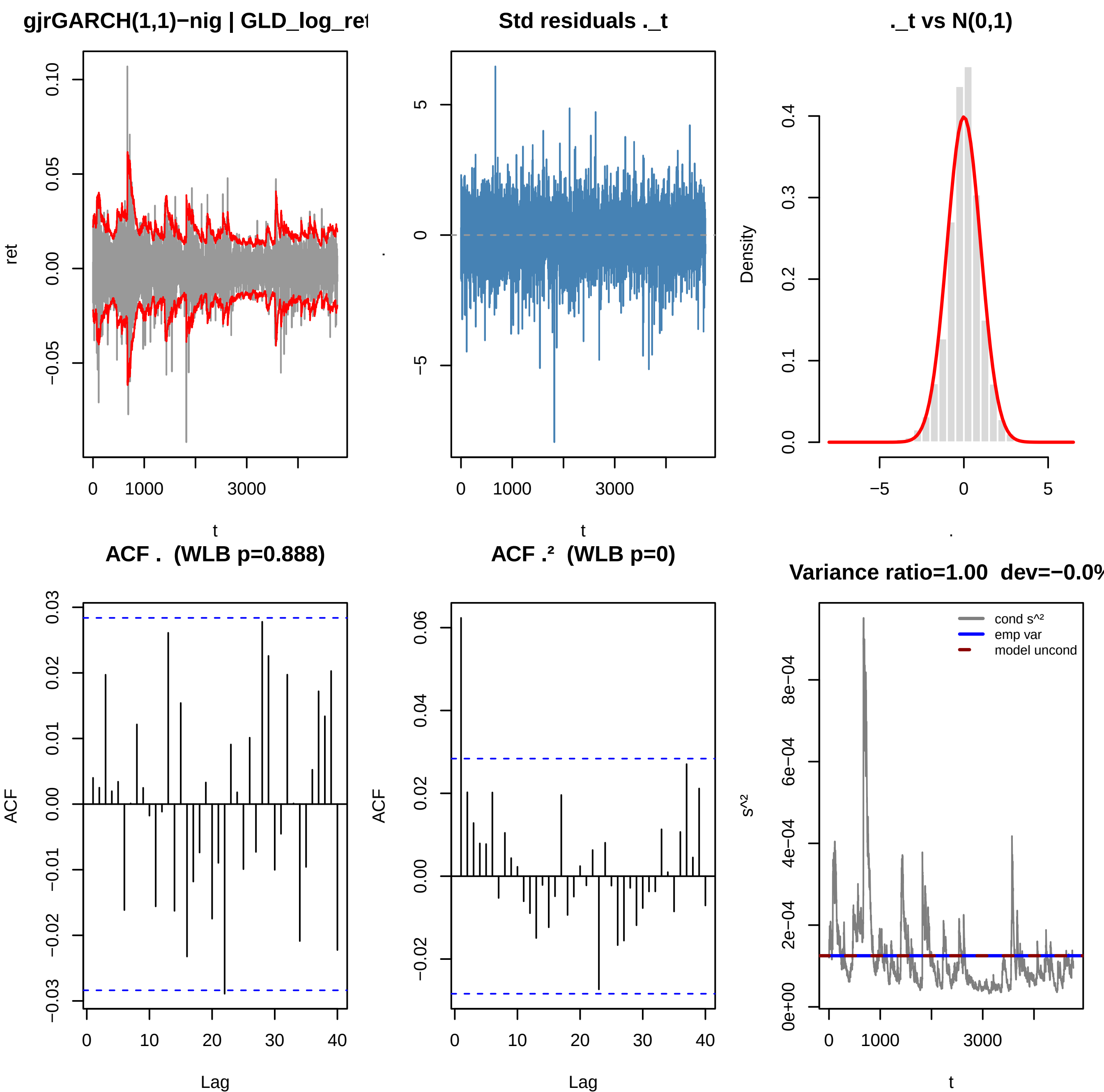

Residuals — GLD_log_return
Squared residuals
ACF residuals
ACF squared (LB p=0)
t
t
Lag
Lag
ACF
ACF
0.10
0.05
0.00
−0.05
−0.10
0.008
0.004
0.000
0.03
0.01
−0.01
−0.03
0.20
0.15
0.10
0.05
0.00
0
1000
2000
3000
4000
0
10
20
30
40

ep 6 — GARCH AIC heatmap (blue=best) — C

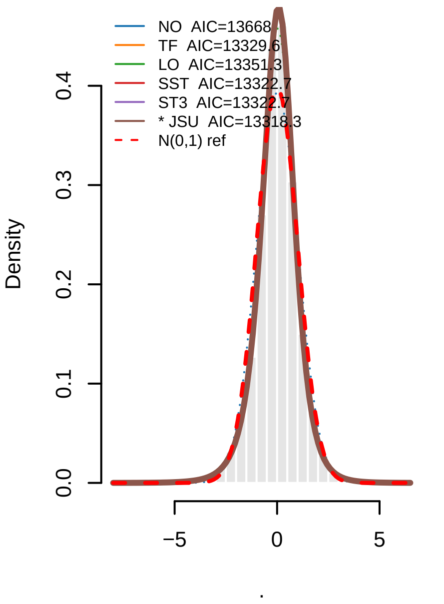


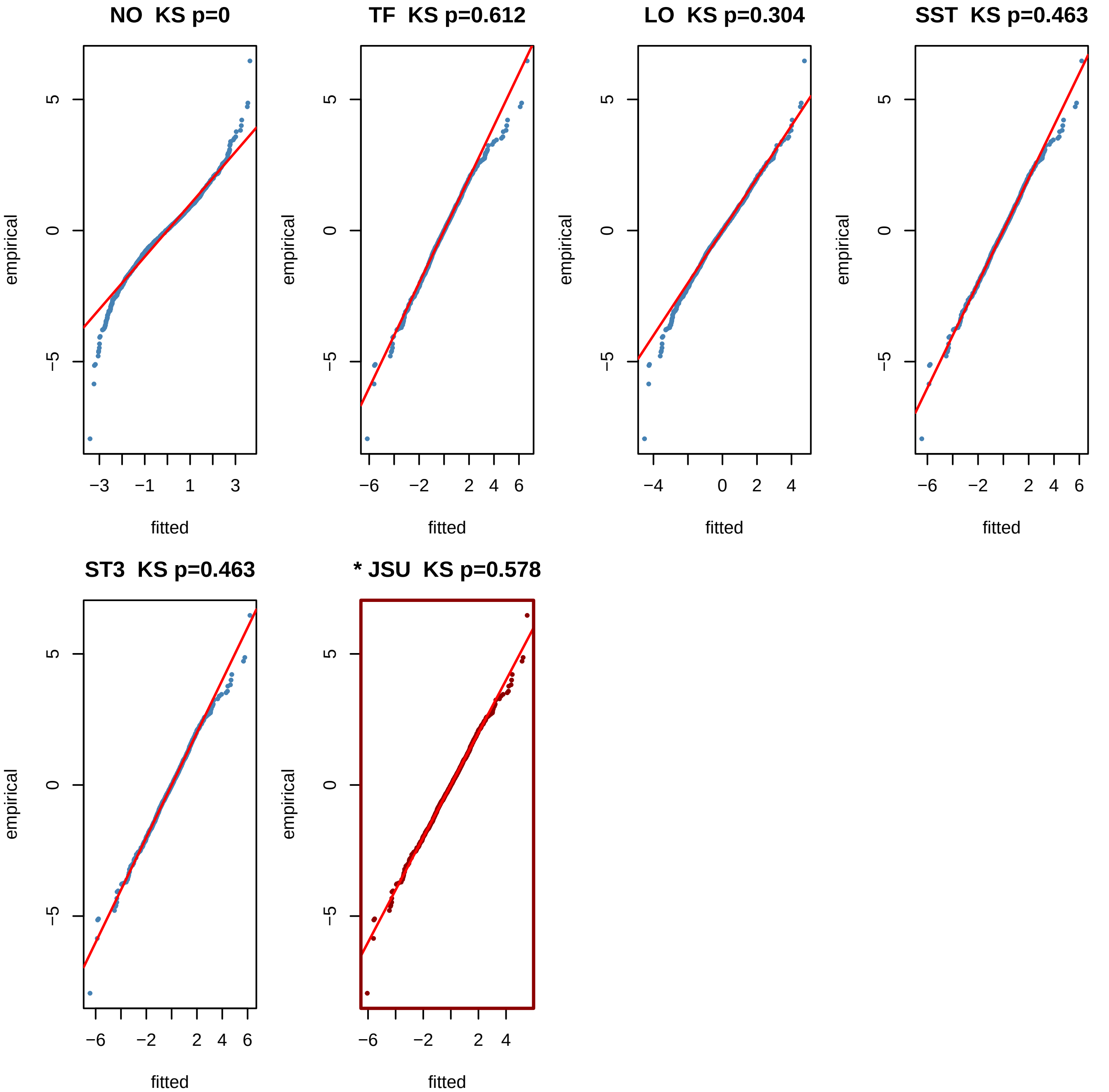


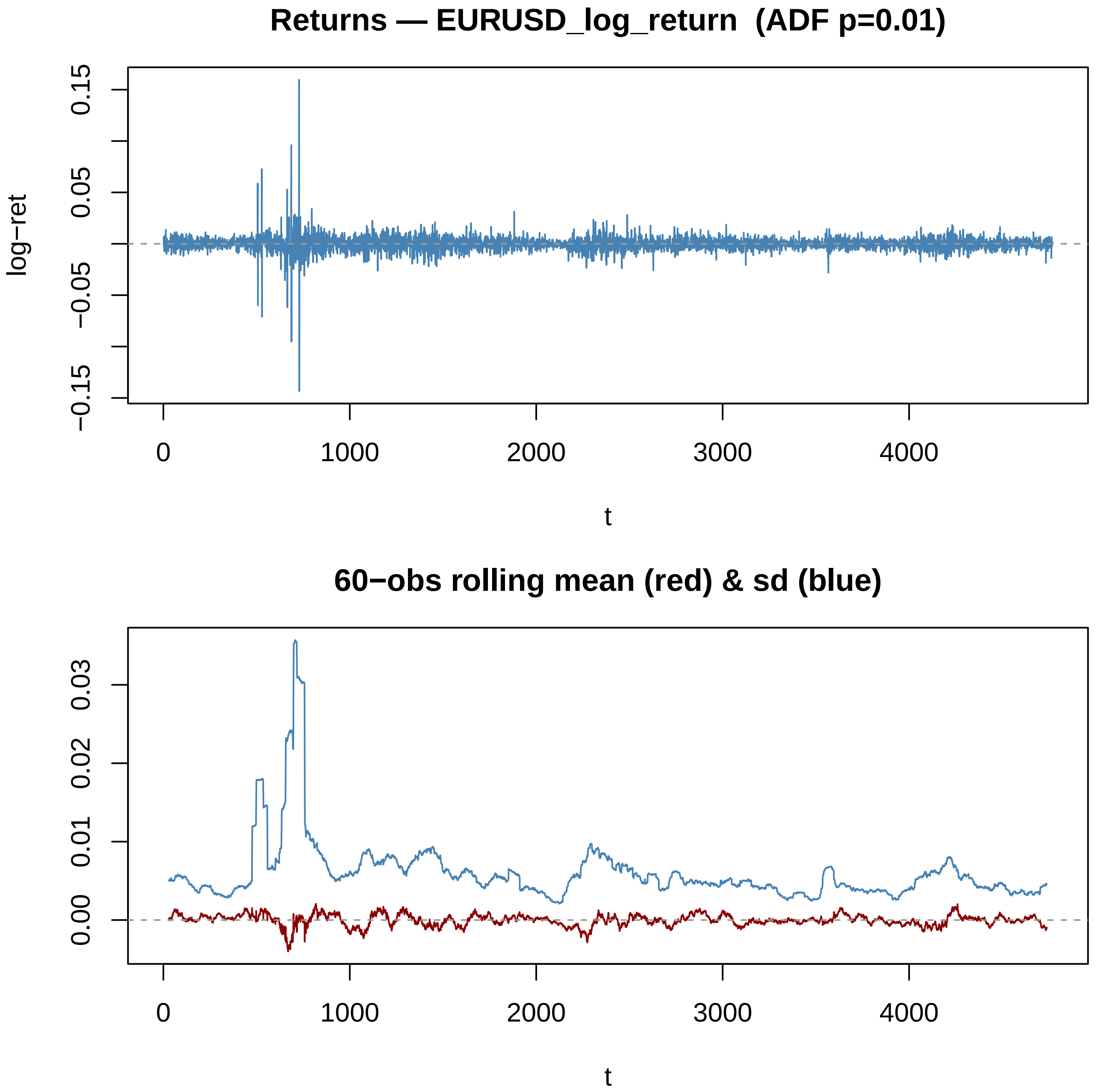
* nig AIC=−6.42
jsu AIC=−6.42
sstd AIC=−6.42
std AIC=−6.41
sged AIC=−6.41
ged AIC=−6.41
norm AIC=−6.34
empirical
theoretical

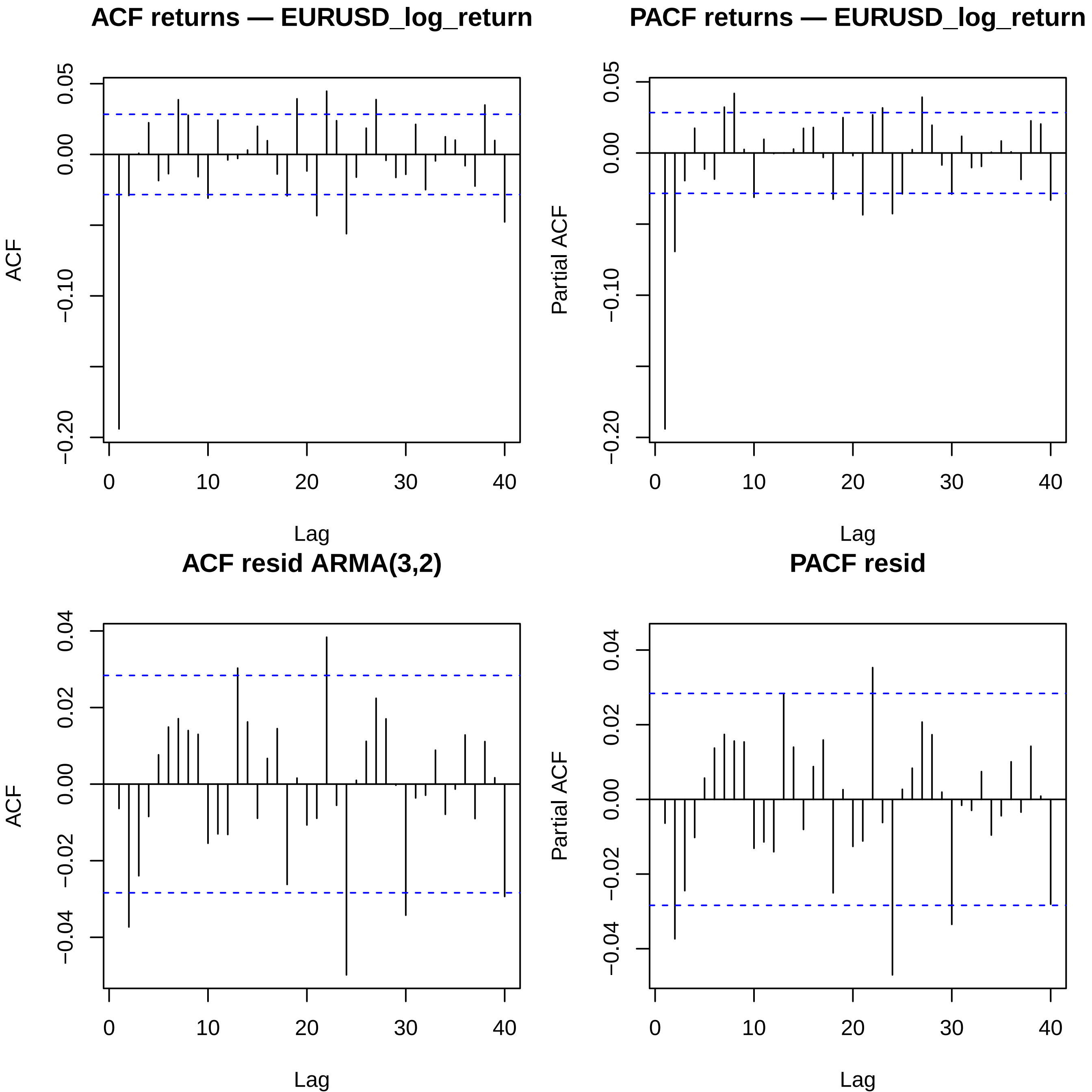

gjrGARCH(1,1)−nig | GLD_log_ret
Std residuals ._t
._t vs N(0,1)
ret
t
Density
ACF . (WLB p=0.888)
ACF .² (WLB p=0)
Variance ratio=1.00 dev=−0.0%
ACF
Lag
s^2
cond s^2
emp var
model uncond

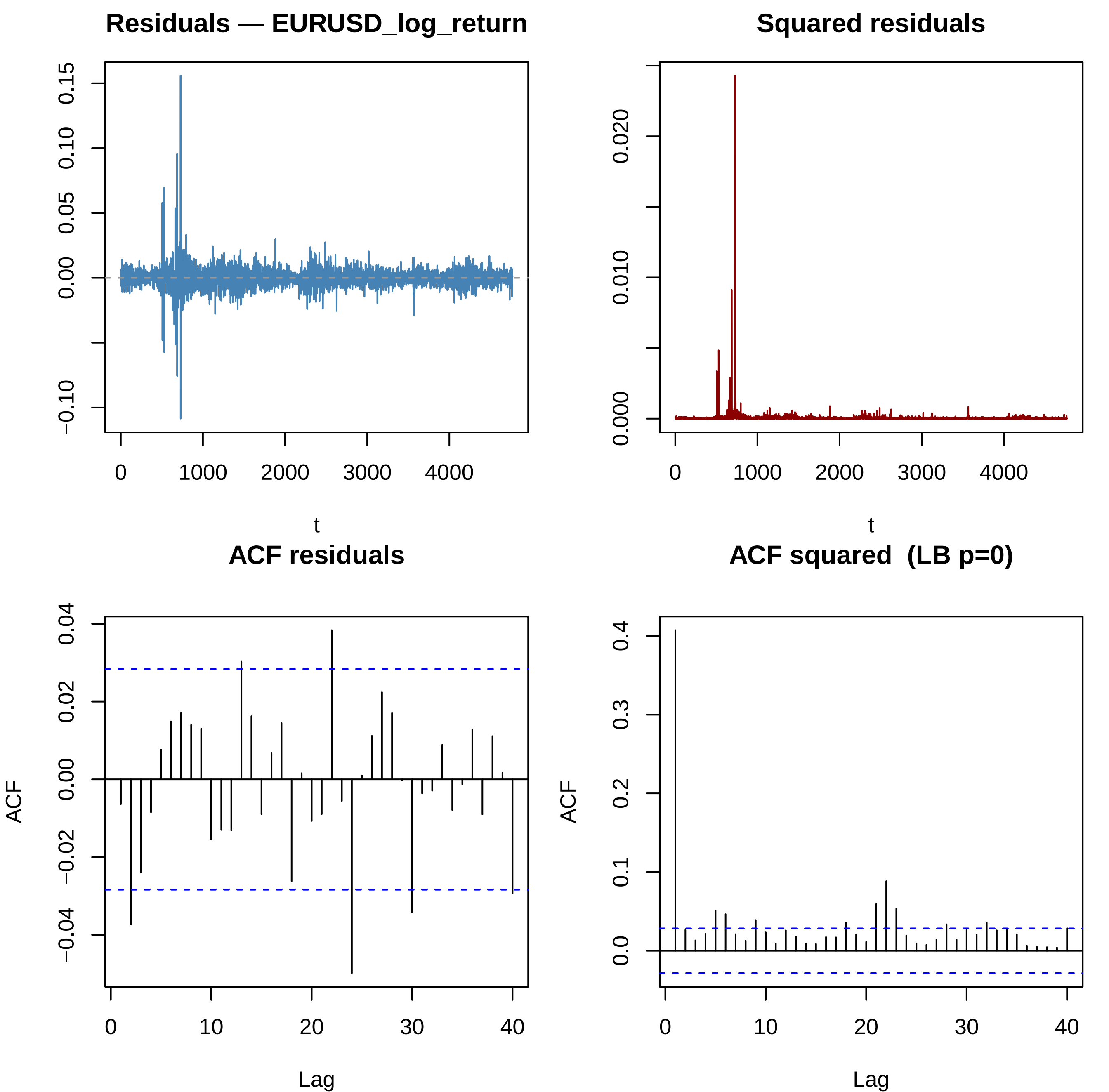
o 8 — PIT density fits to ._t — GLD_l




NO KS p=0
TF KS p=0.612
LO KS p=0.304
SST KS p=0.463
ST3 KS p=0.463
* JSU KS p=0.578
empirical
fitted
−5
0
5
−3
−1
1
3
−6
−2
2
4
6
−4
0

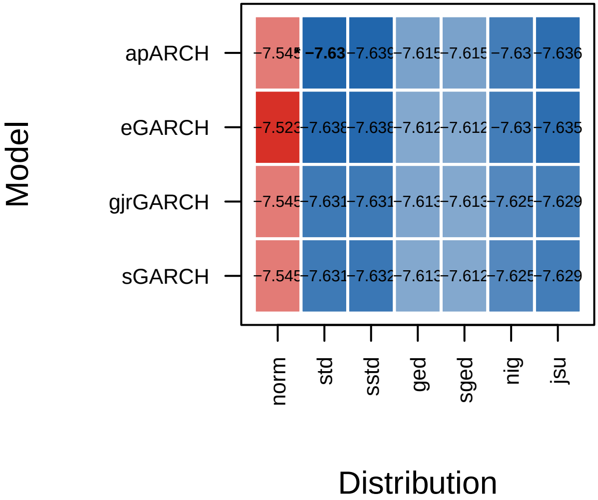

Returns — EURUSD_log_return (ADF p=0.01)
log−ret
0.15
0.05
−0.05
−0.15
0
1000
2000
3000
4000
t
60−obs rolling mean (red) & sd (blue)
0.03
0.02
0.01
0.00
0
1000
2000
3000
4000
t

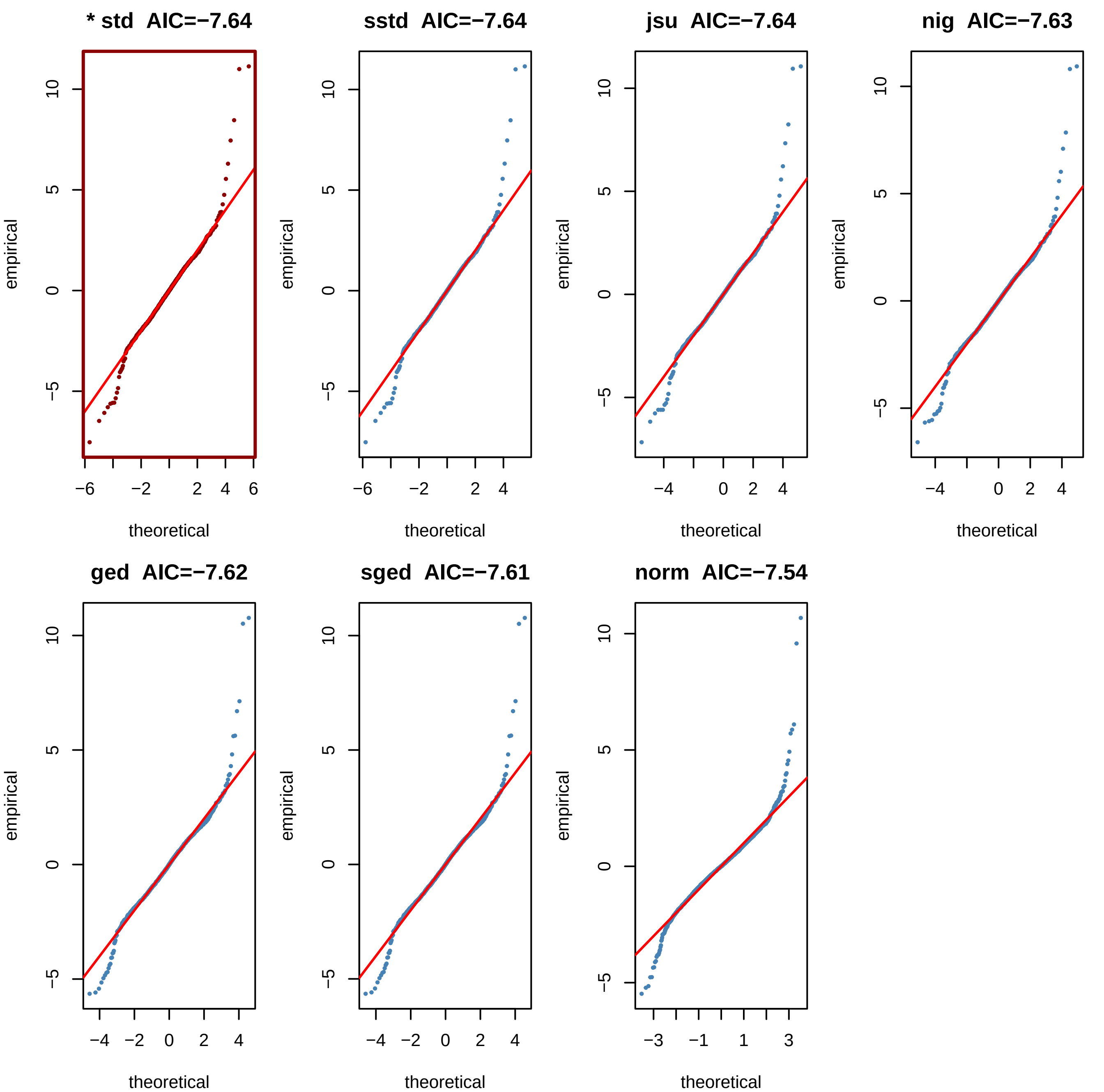

ACF returns — EURUSD_log_return
PACF returns — EURUSD_log_return
ACF resid ARMA(3,2)
PACF resid
ACF
Partial ACF
Lag
0.05
0.00
−0.10
−0.20
0.04
0.02
−0.02
−0.04
0
10
20
30
40



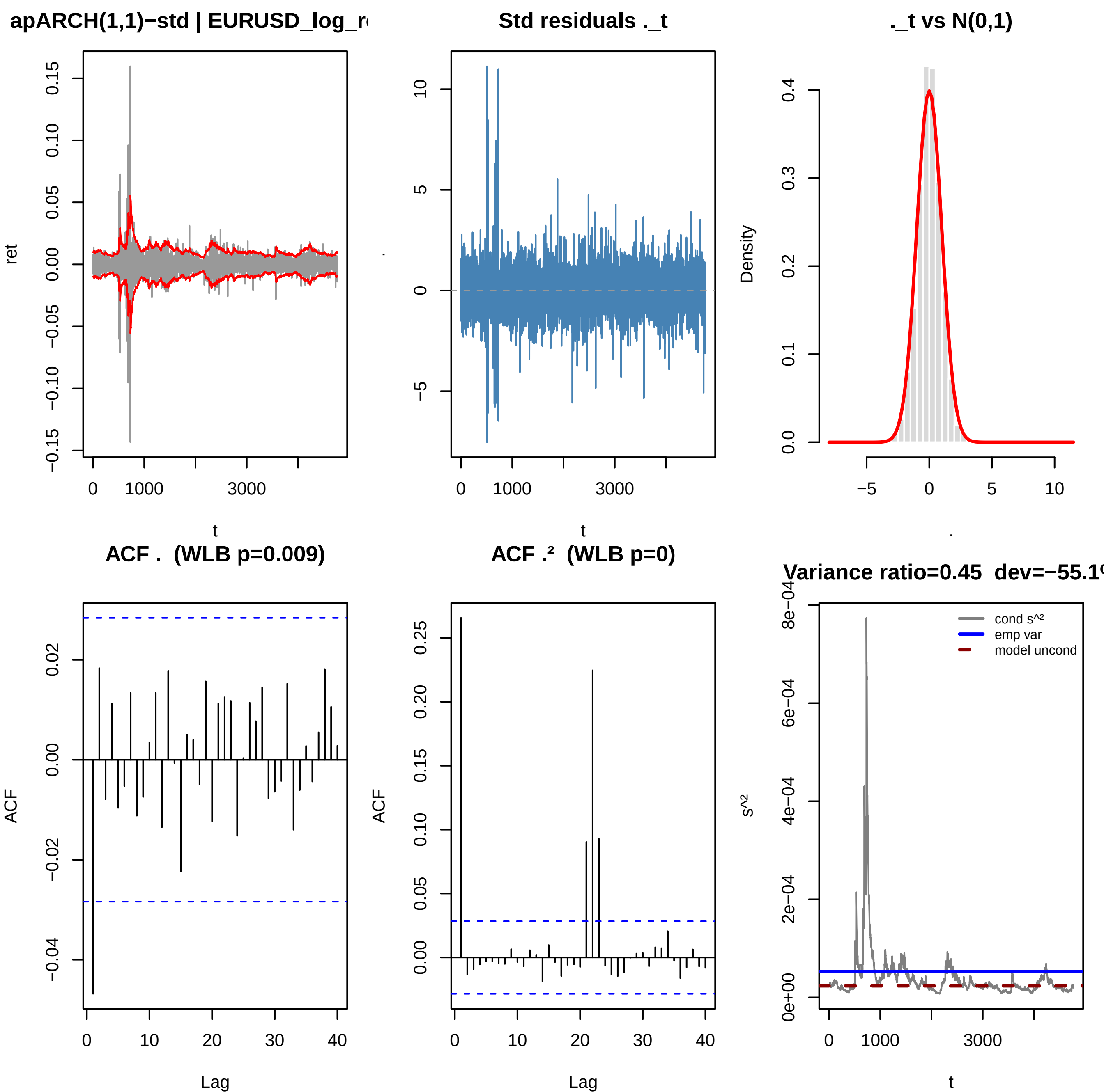
Residuals — EURUSD_log_return
0.15
0.10
0.05
0.00
−0.10
0
1000
2000
3000
4000
t
Squared residuals
0.020
0.010
0.000
0
1000
2000
3000
4000
t
ACF residuals
ACF
0.04
0.02
0.00
−0.02
−0.04
0
10
20
30
40
Lag
ACF squared (LB p=0)
ACF
0.4
0.3
0.2
0.1
0.0
0
10
20
30
40
Lag

6 — GARCH AIC heatmap (blue=best) — EU



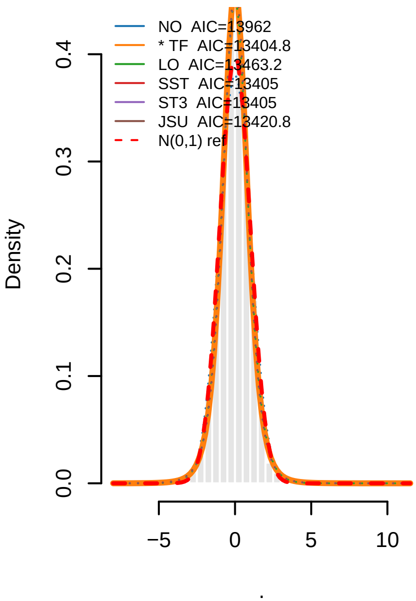


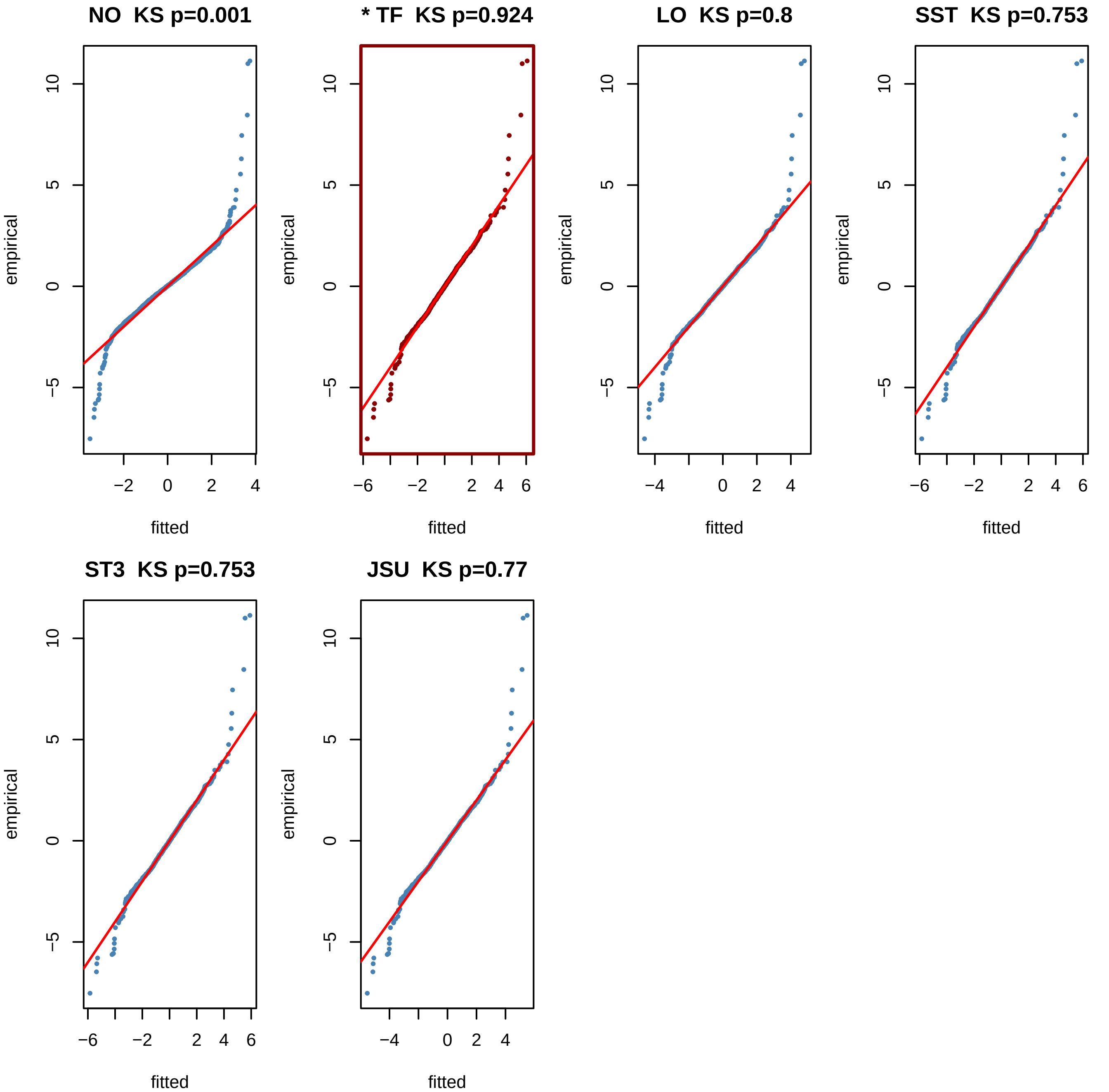
* std AIC=−7.64
sstd AIC=−7.64
jsu AIC=−7.64
nig AIC=−7.63
ged AIC=−7.62
sged AIC=−7.61
norm AIC=−7.54
empirical
theoretical



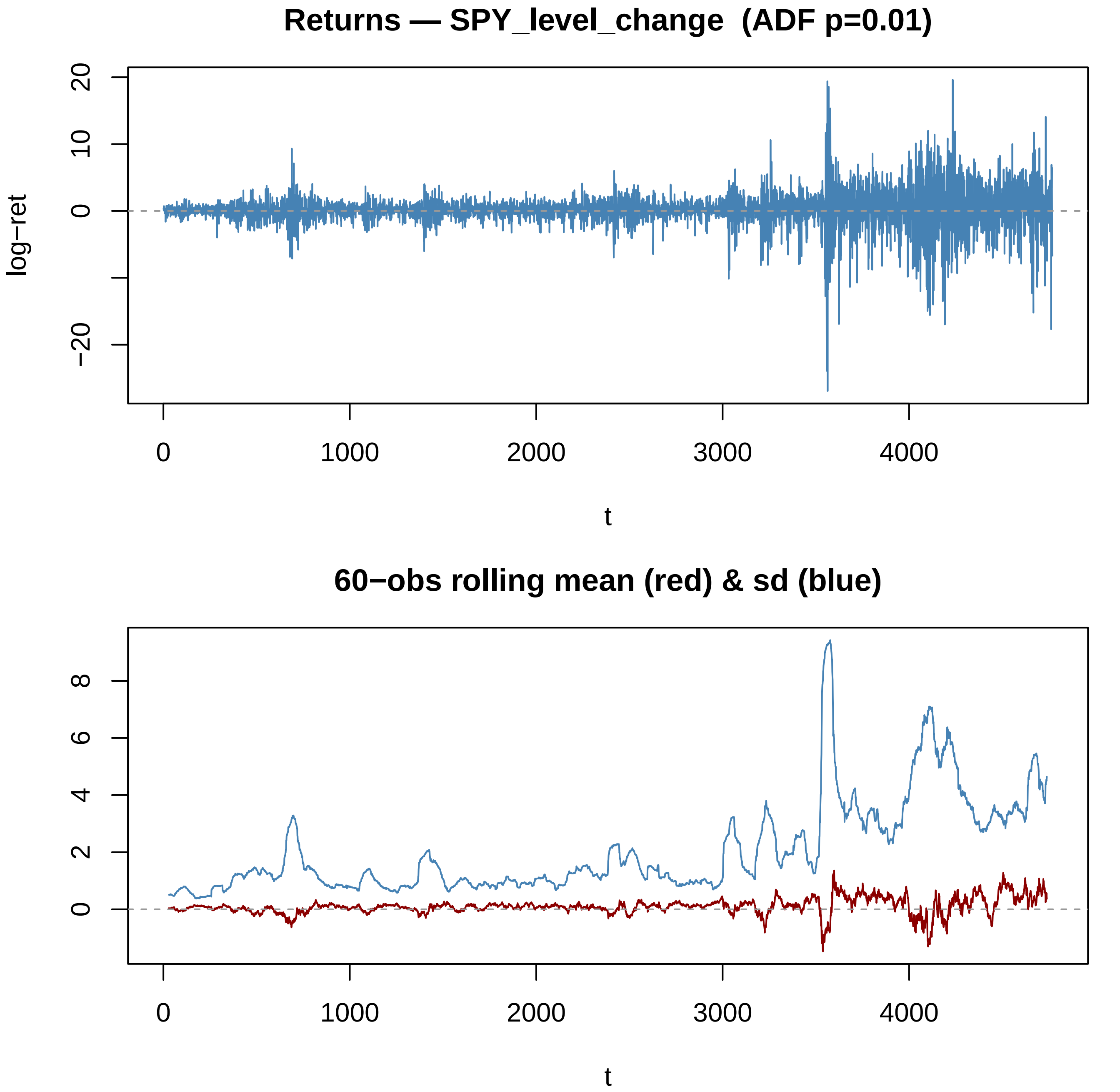
apARCH(1,1)−std | EURUSD_log_r
ret
t
Std residuals ._t
t
._t vs N(0,1)
Density
ACF . (WLB p=0.009)
ACF
Lag
ACF .² (WLB p=0)
ACF
Lag
Variance ratio=0.45 dev=−55.1
cond s^2
emp var
model uncond
s^2
t

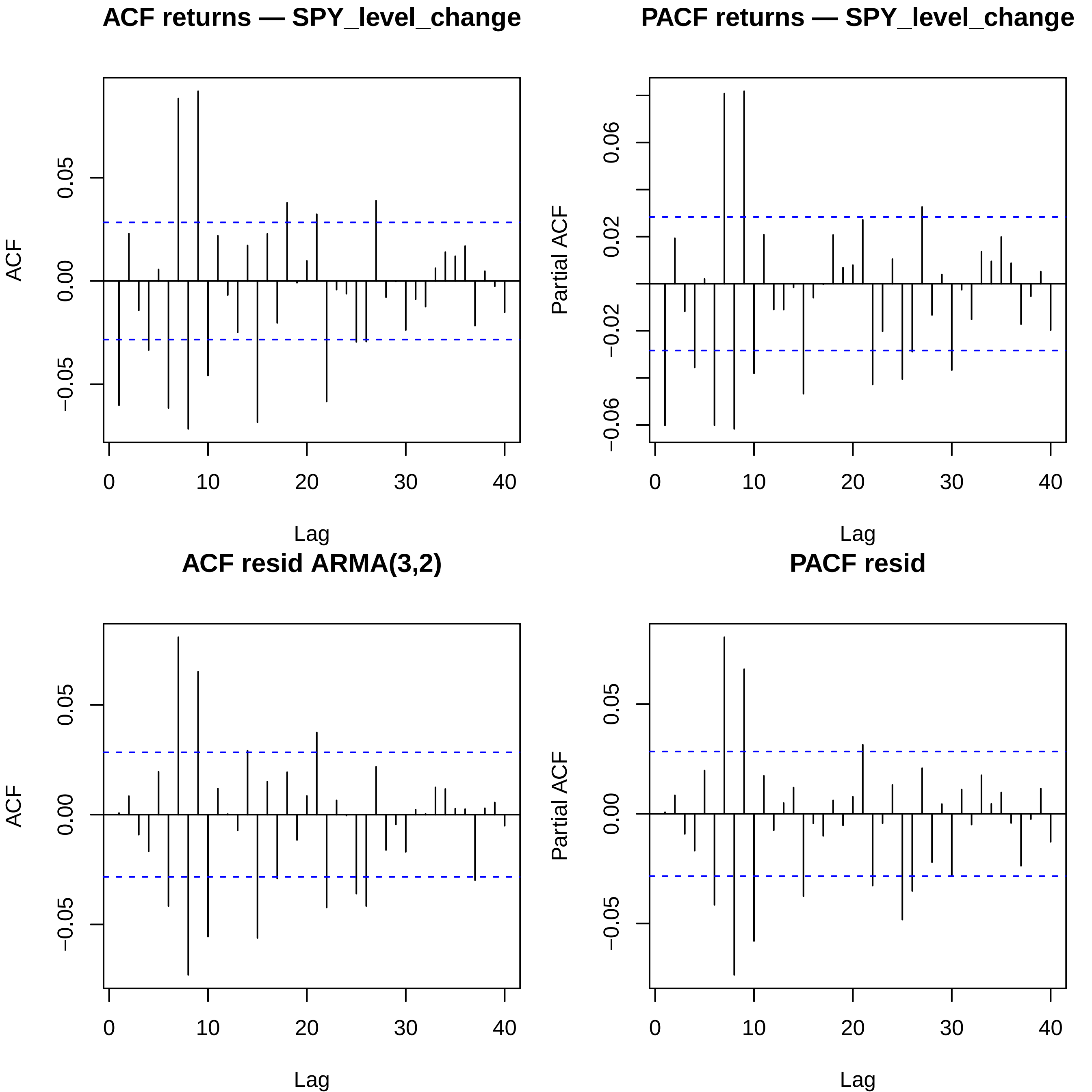
— PIT density fits to ._t — EURUSD



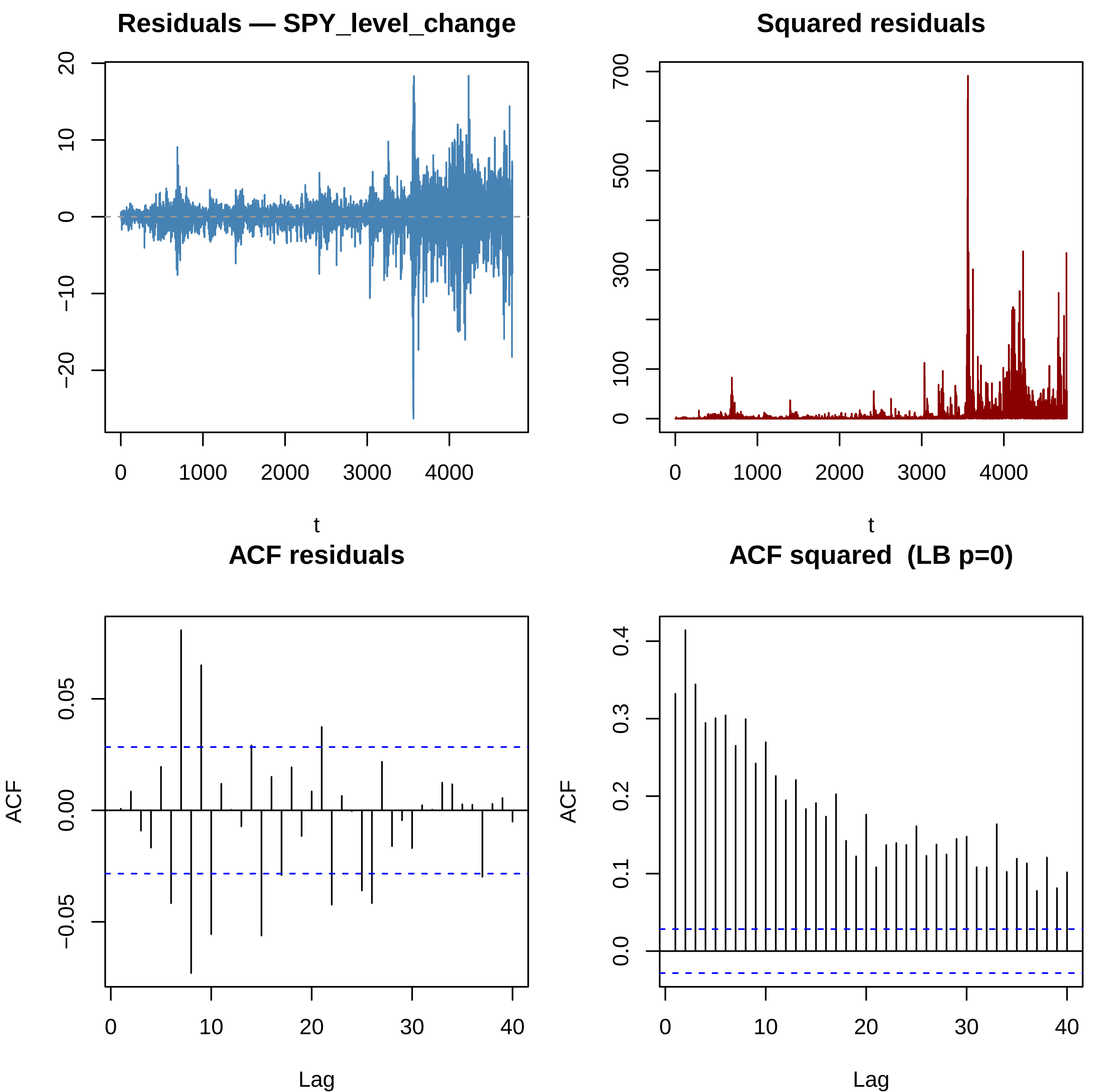


NO KS p=0.001
* TF KS p=0.924
LO KS p=0.8
SST KS p=0.753
ST3 KS p=0.753
JSU KS p=0.77
empirical
fitted

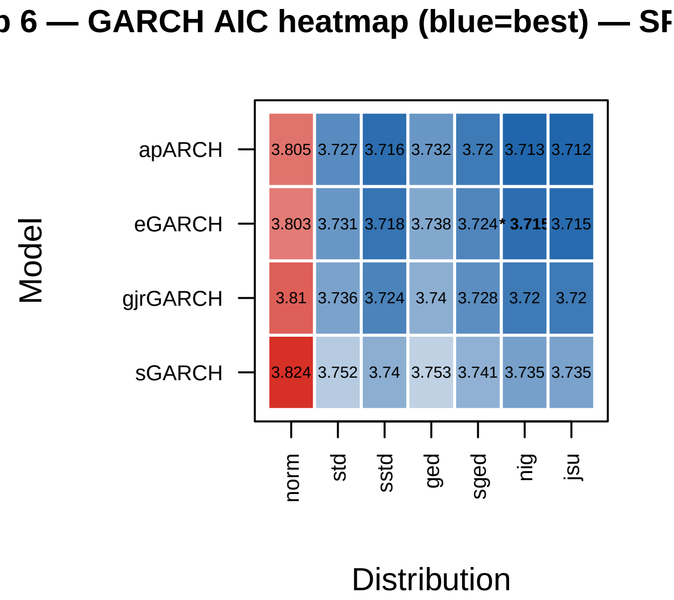

Returns — SPY_level_change (ADF p=0.01)
log−ret
20
10
0
−20
0
1000
2000
3000
4000
t
60−obs rolling mean (red) & sd (blue)
8
6
4
2
0
0
1000
2000
3000
4000
t

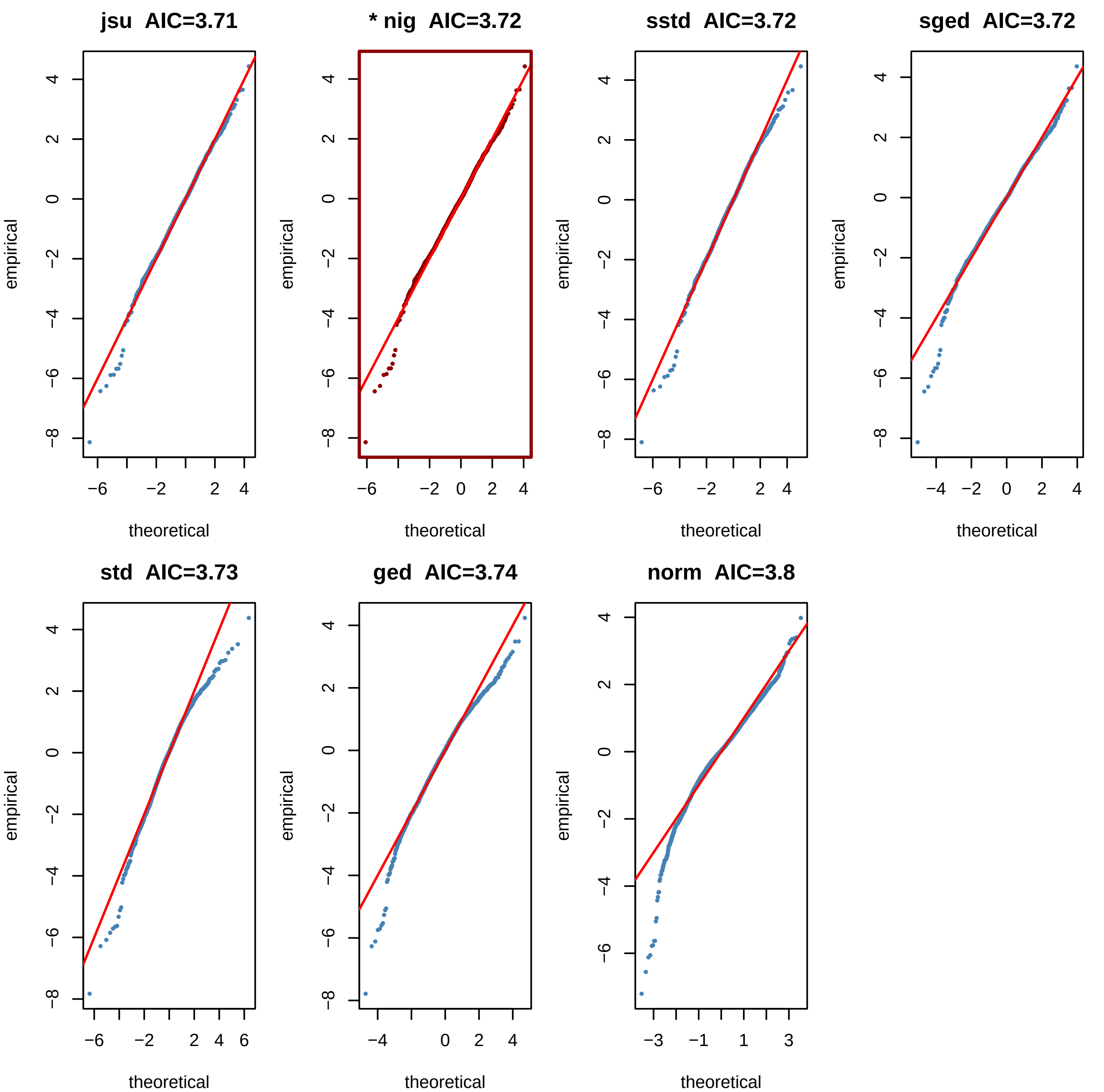


ACF returns — SPY_level_change
PACF returns — SPY_level_change
ACF resid ARMA(3,2)
PACF resid
ACF
Partial ACF
Lag
0
10
20
30
40
0.05
0.00
−0.05
0.06
0.02
−0.02
−0.06

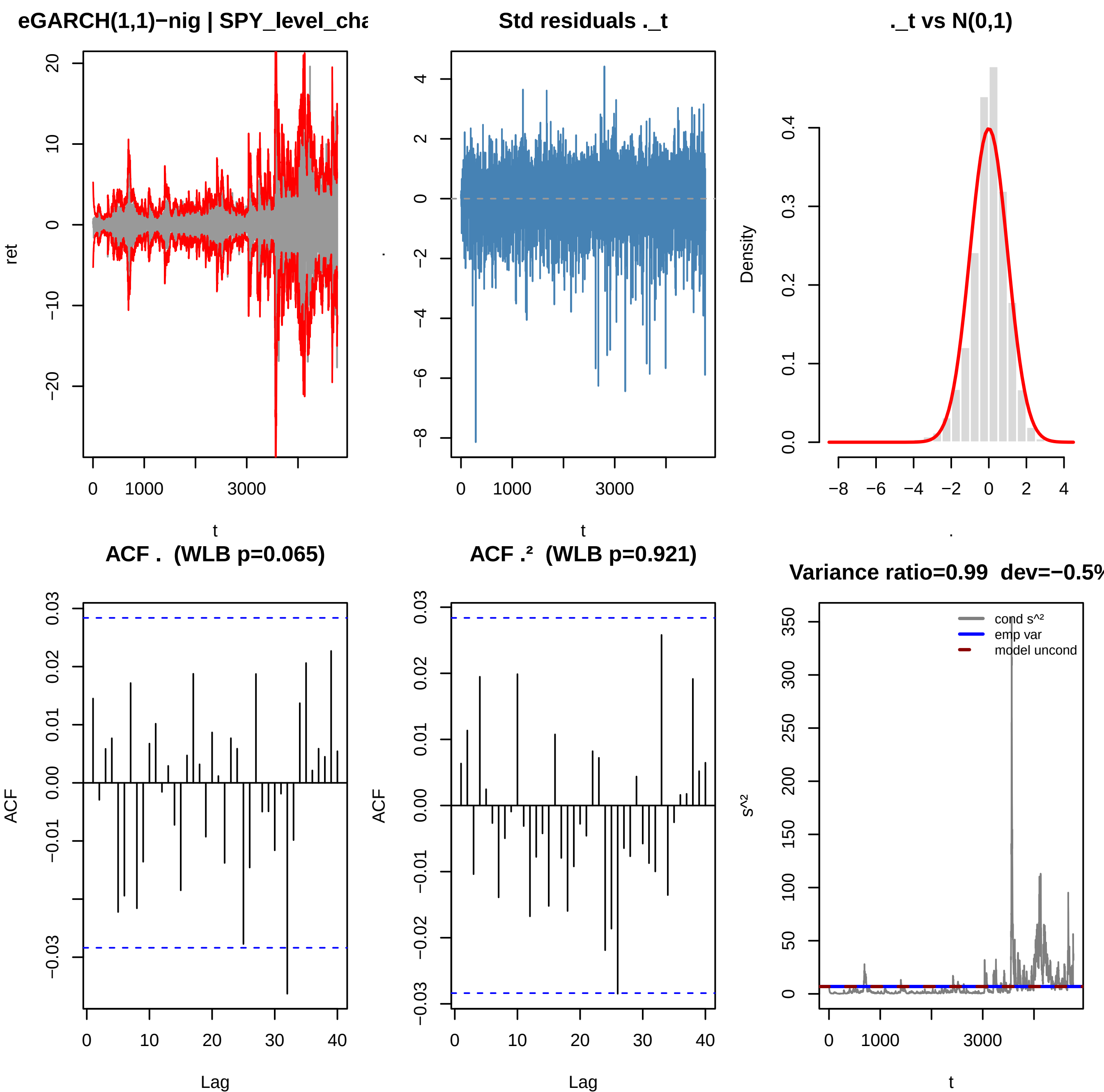


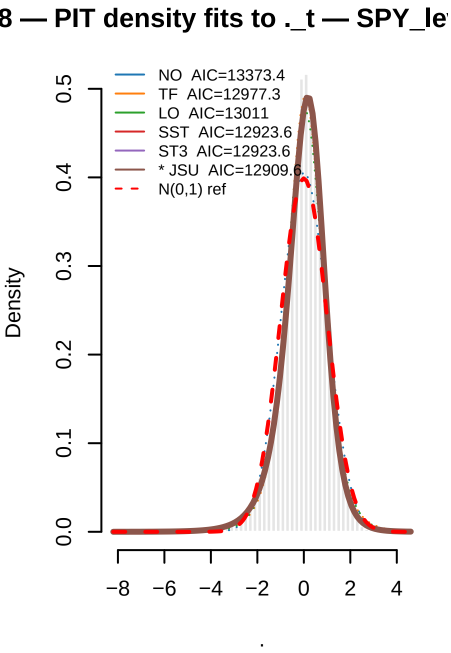
Residuals — SPY_level_change
Squared residuals
ACF residuals
ACF squared (LB p=0)
t
t
Lag
Lag
ACF
ACF

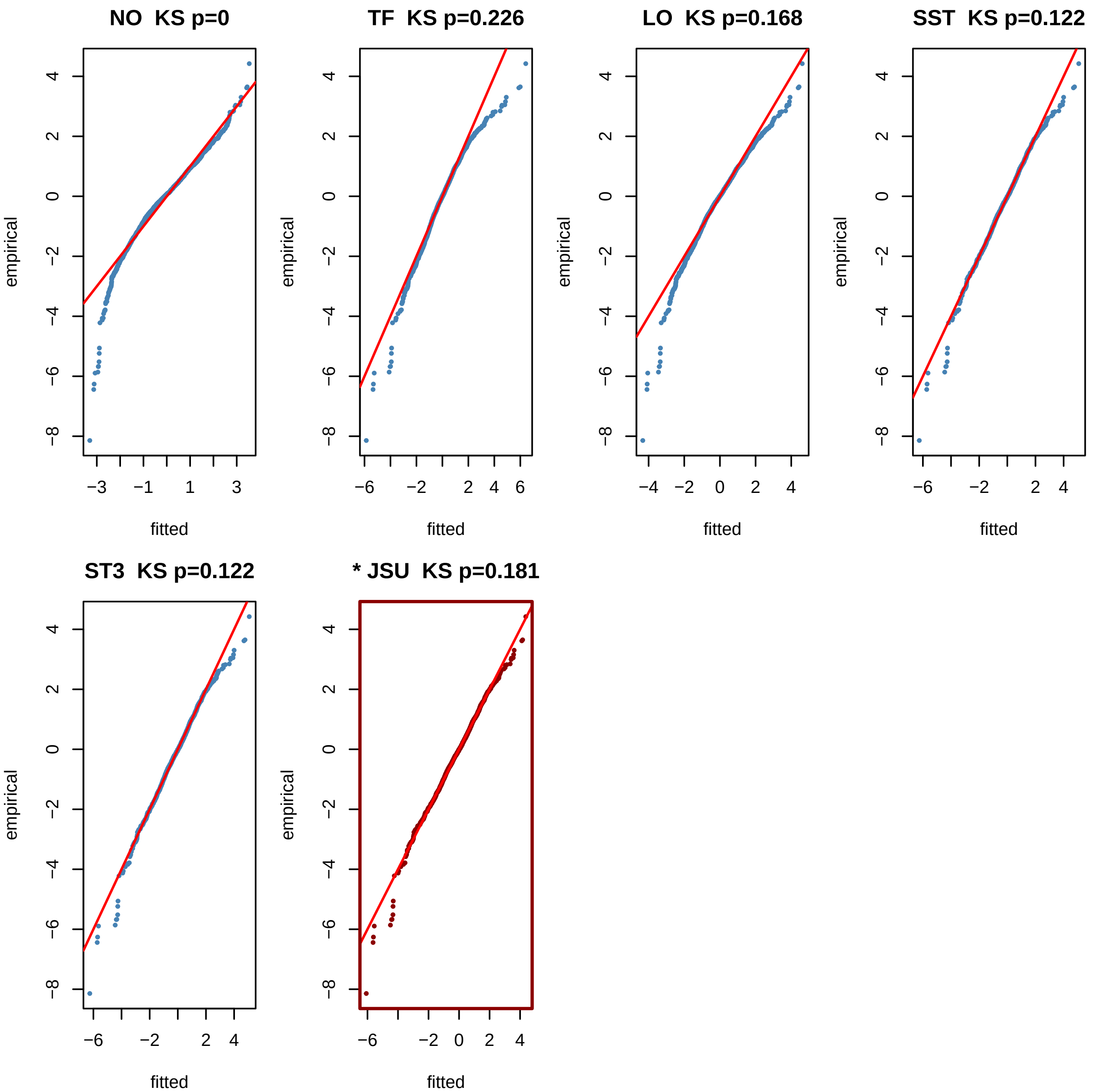

b 6 — GARCH AIC heatmap (blue=best) — SF
Model
apARCH
eGARCH
gjrGARCH
sGARCH
3.805
3.727
3.716
3.732
3.72
3.713
3.712
3.803
3.731
3.718
3.738
3.724
* 3.715
3.715
3.81
3.736
3.724
3.74
3.728
3.72
3.72
3.824
3.752
3.74
3.753
3.741
3.735
3.735
norm
std
sstd
ged
sged
nig
jsu
Distribution



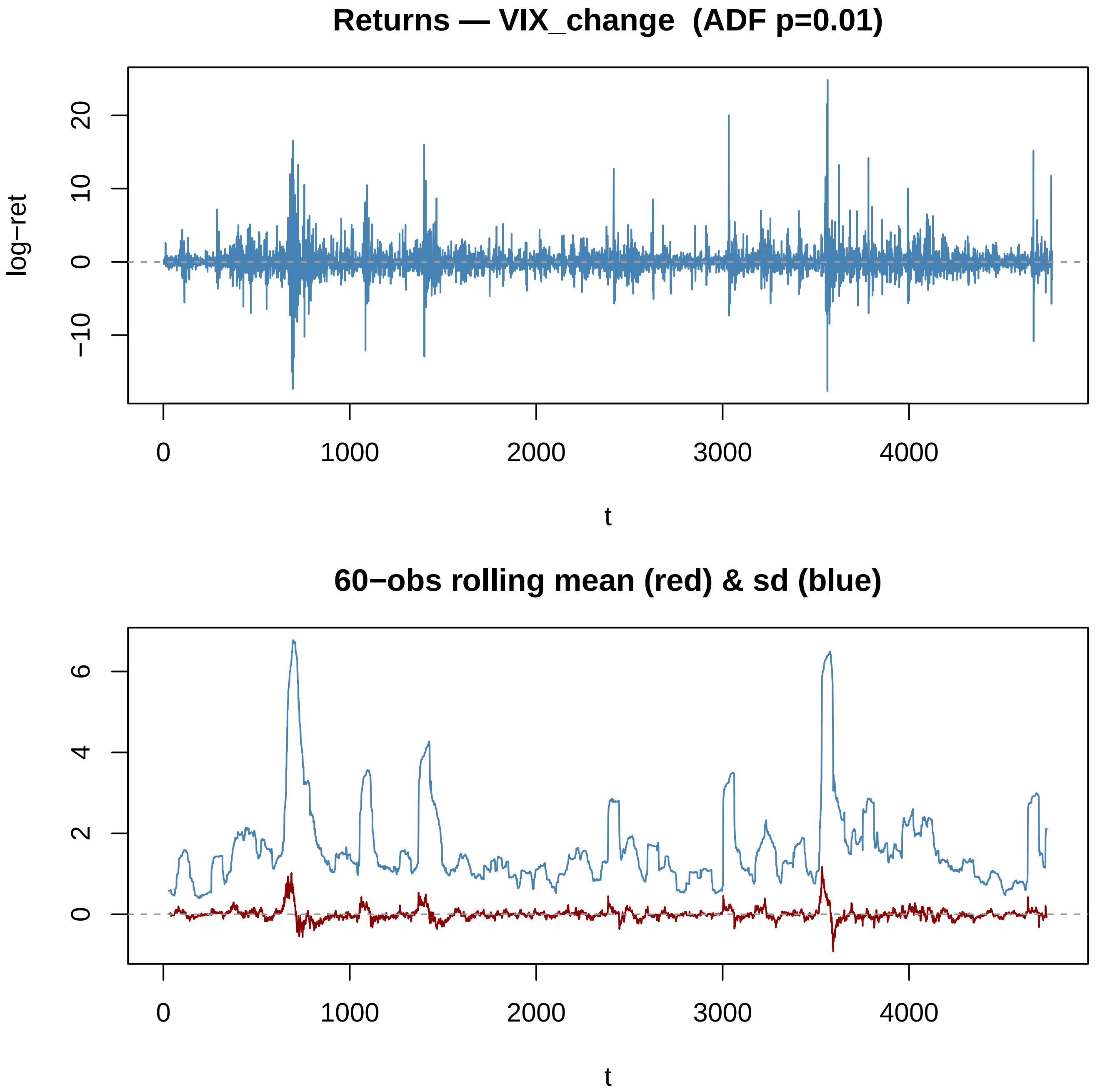
jsu AIC=3.71
* nig AIC=3.72
sstd AIC=3.72
sged AIC=3.72
std AIC=3.73
ged AIC=3.74
norm AIC=3.8
empirical
theoretical

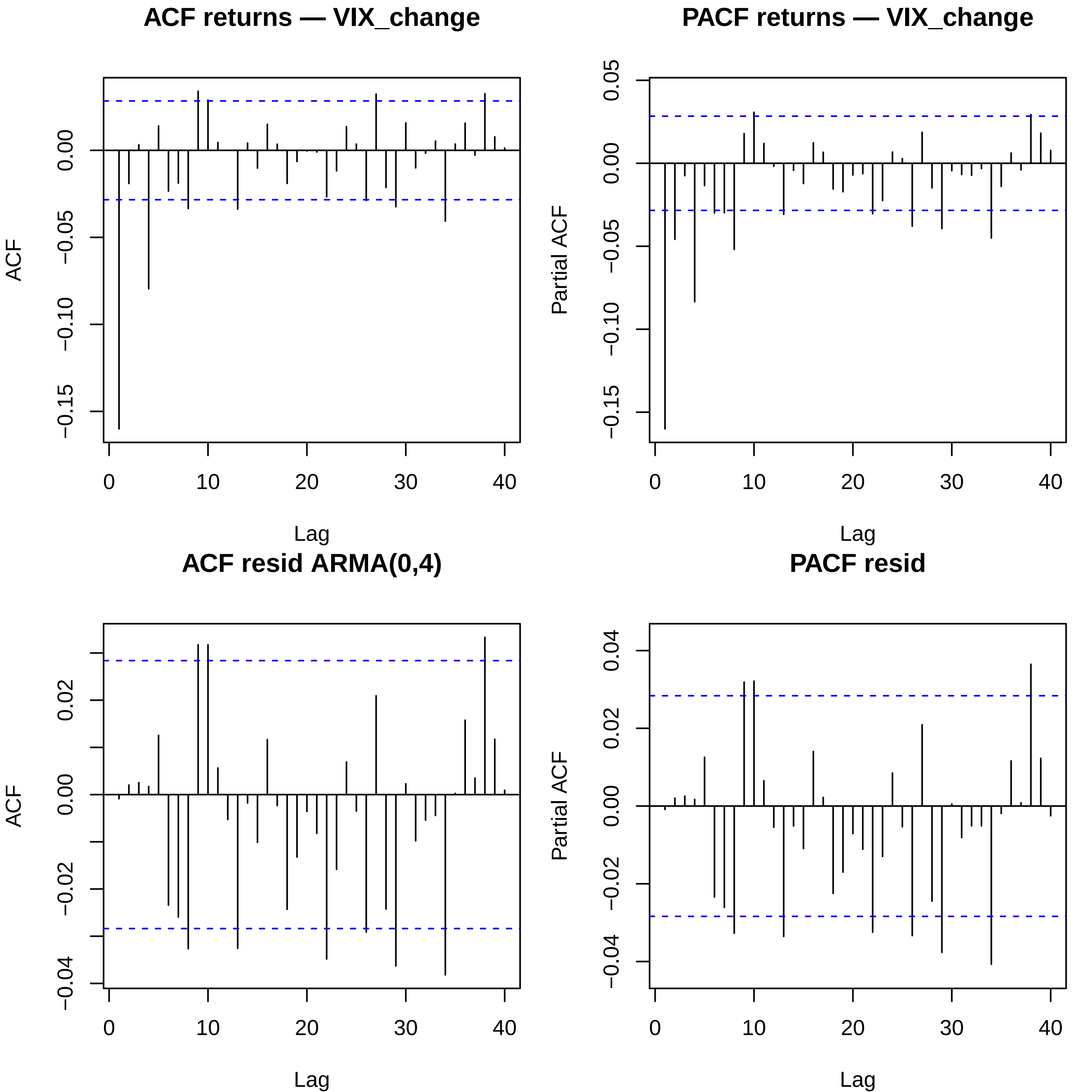

eGARCH(1,1)−nig | SPY_level_cha
ret
t
Std residuals ._t
t
._t vs N(0,1)
Density
ACF . (WLB p=0.065)
ACF
Lag
ACF .² (WLB p=0.921)
ACF
Lag
Variance ratio=0.99 dev=−0.5%
s^2
t
cond s^2
emp var
model uncond

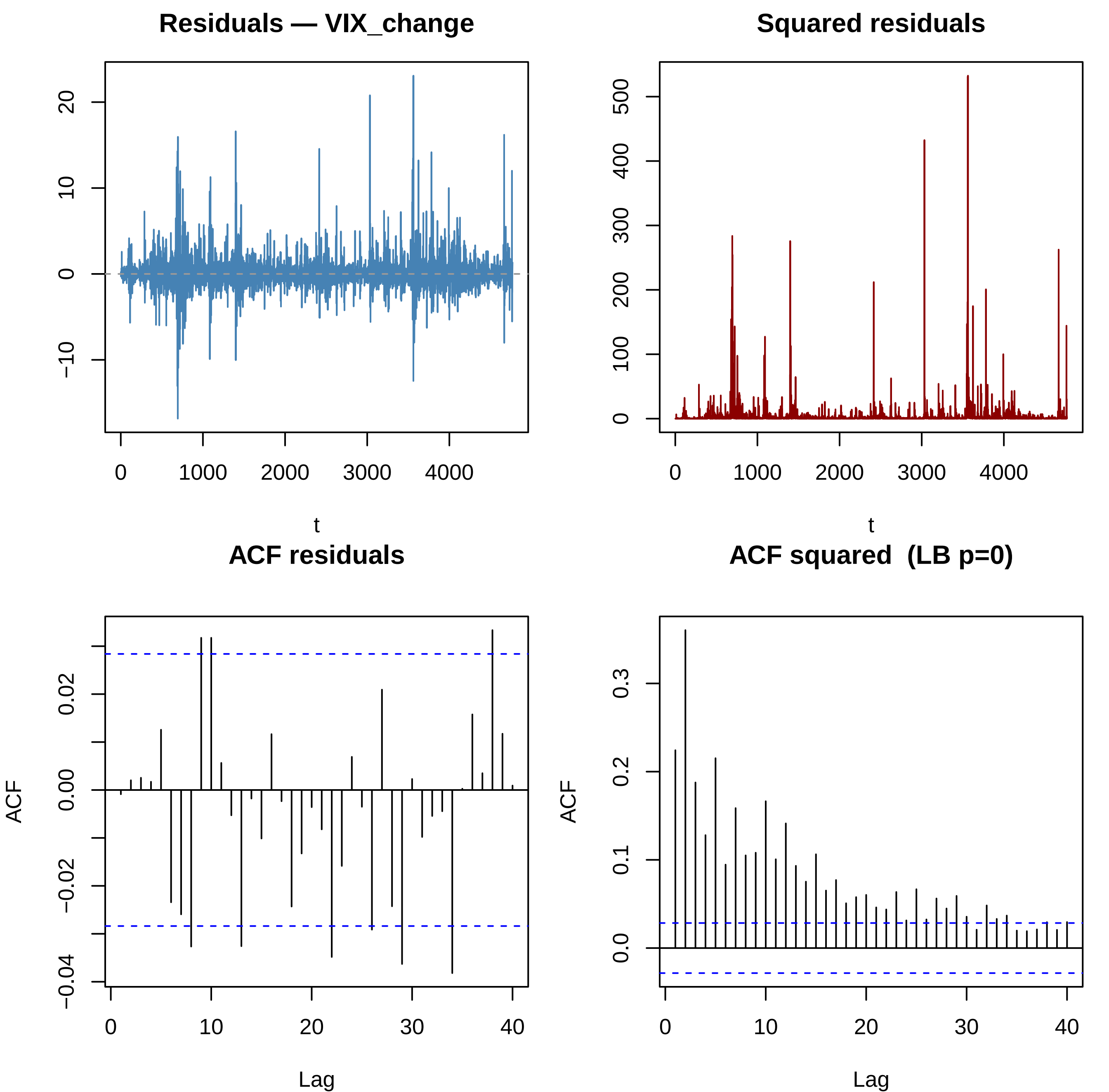

8 — PIT density fits to ._t — SPY_le
NO AIC=13373.4
TF AIC=12977.3
LO AIC=13011
SST AIC=12923.6
ST3 AIC=12923.6
* JSU AIC=12909.6
N(0,1) ref
Density
0.0
0.1
0.2
0.3
0.4
0.5
−8
−6
−4
−2
0
2
4
.



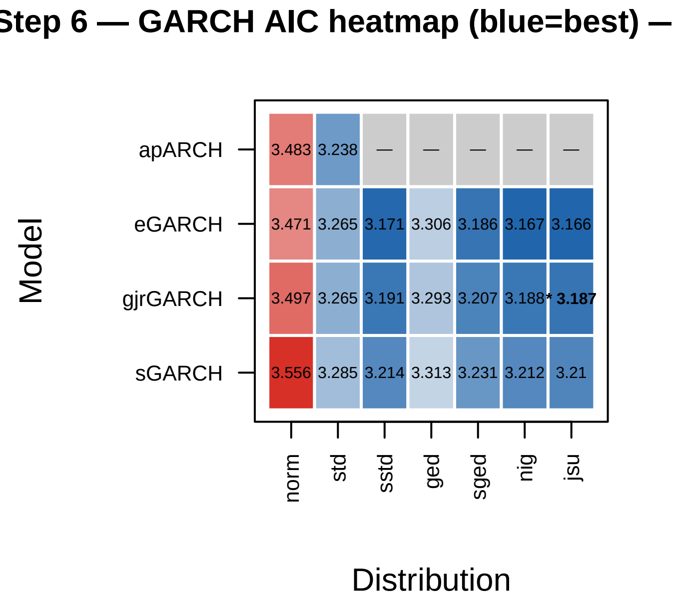
NO KS p=0
TF KS p=0.226
LO KS p=0.168
SST KS p=0.122
ST3 KS p=0.122
* JSU KS p=0.181
empirical
fitted
4
2
0
−2
−4
−6
−8
−3
−1
1
3
−6
−2
2
4
6
−4
0

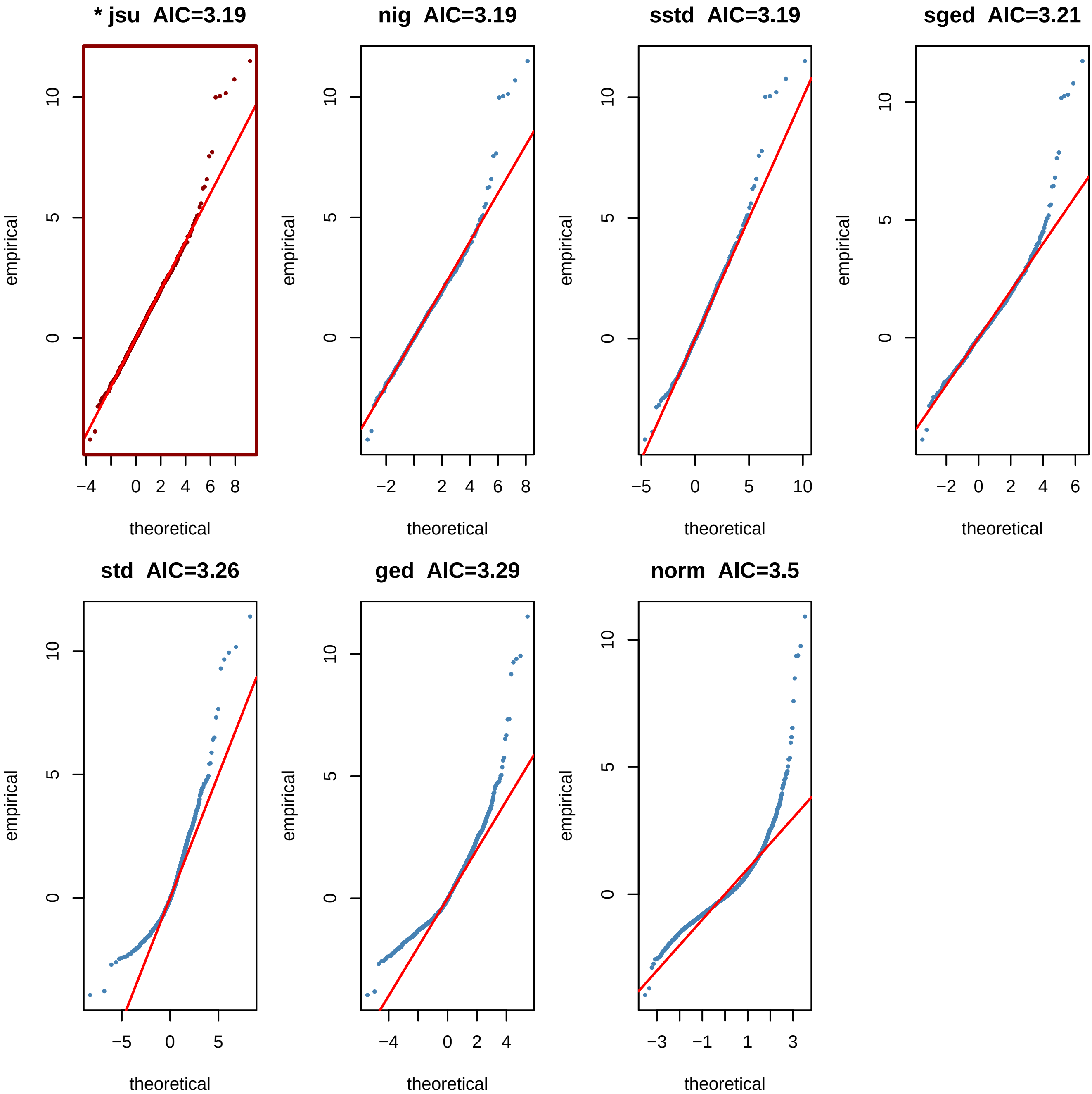

Returns — VIX_change (ADF p=0.01)
log−ret
20
10
0
−10
0
1000
2000
3000
4000
t
60−obs rolling mean (red) & sd (blue)
6
4
2
0
0
1000
2000
3000
4000
t

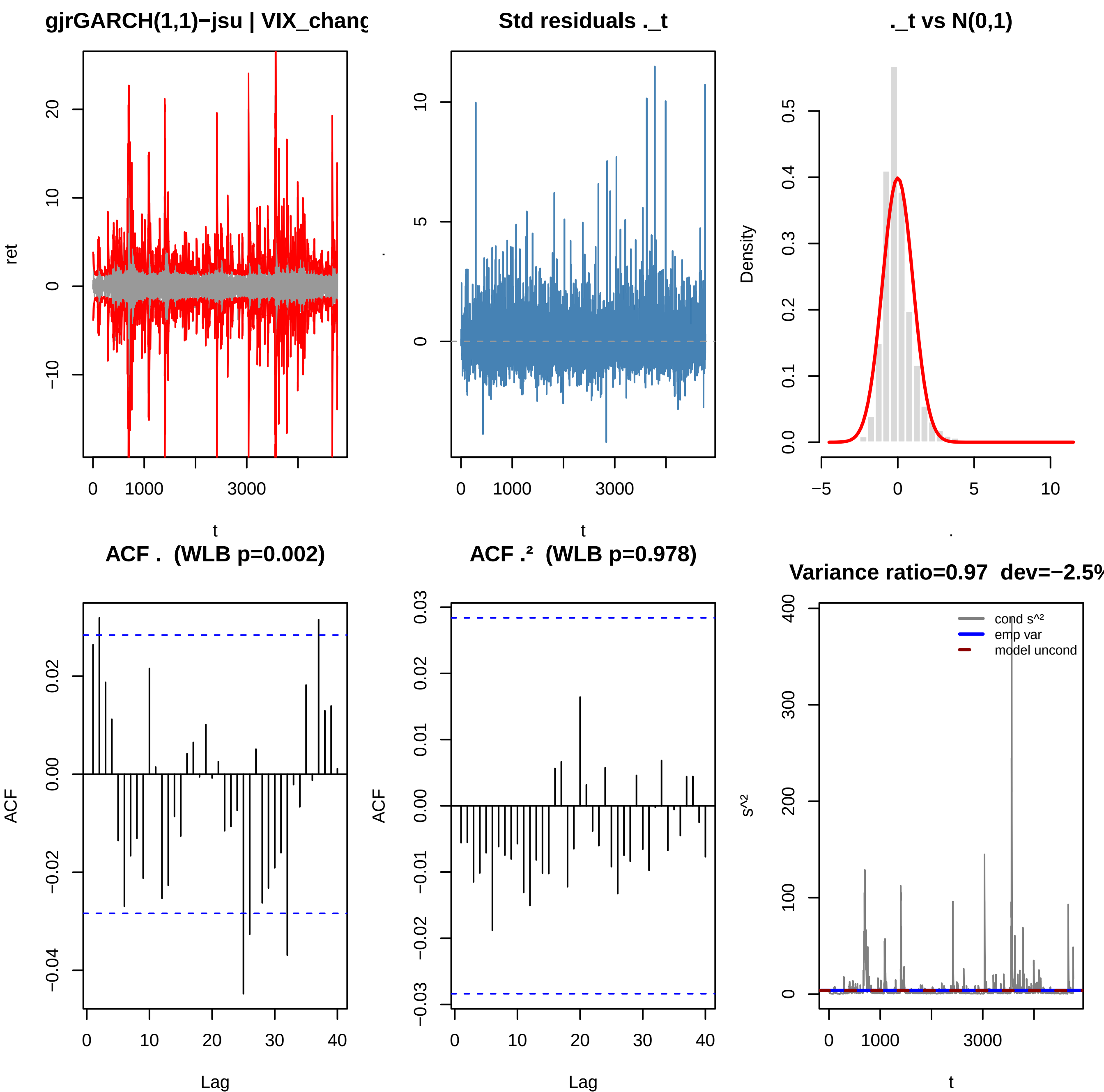

ACF returns — VIX_change
ACF
0.00
−0.05
−0.10
−0.15
0
10
20
30
40
Lag
PACF returns — VIX_change
Partial ACF
0.05
0.00
−0.05
−0.10
−0.15
0
10
20
30
40
Lag
ACF resid ARMA(0,4)
ACF
0.02
0.00
−0.02
−0.04
0
10
20
30
40
Lag
PACF resid
Partial ACF
0.04
0.02
0.00
−0.02
−0.04
0
10
20
30
40
Lag



Residuals — VIX_change
Squared residuals
ACF residuals
ACF squared (LB p=0)
t
Lag
ACF

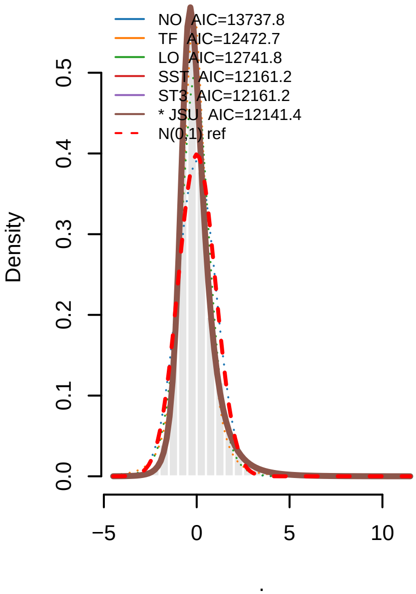

Step 6 — GARCH AIC heatmap (blue=best) —
Model
apARCH
eGARCH
gjrGARCH
sGARCH
3.483 3.238 — — — — —
3.471 3.265 3.171 3.306 3.186 3.167 3.166
3.497 3.265 3.191 3.293 3.207 3.188 * 3.187
3.556 3.285 3.214 3.313 3.231 3.212 3.21
norm
std
sstd
ged
sged
nig
jsu
Distribution

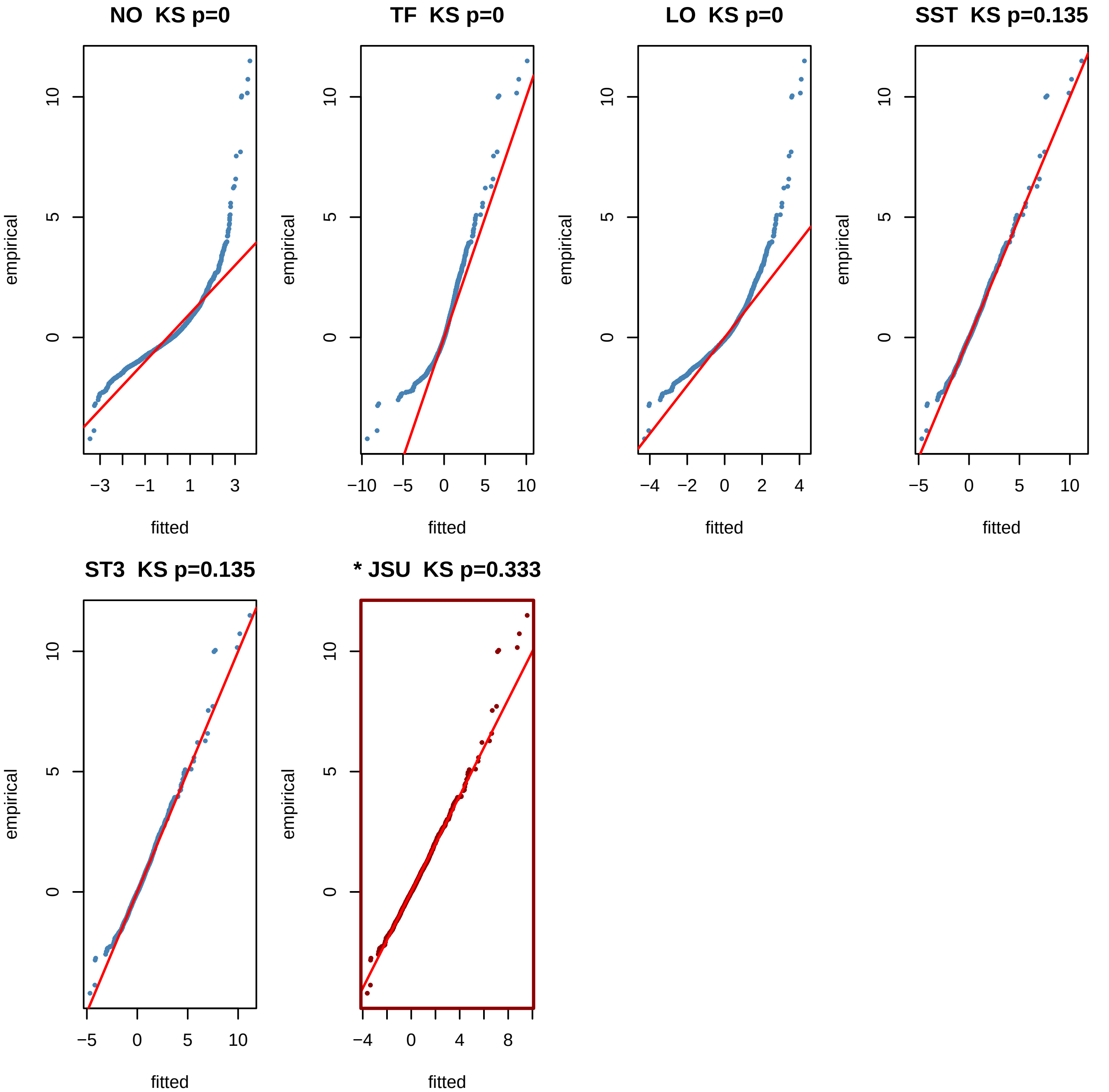

* jsu AIC=3.19
nig AIC=3.19
sstd AIC=3.19
sged AIC=3.21
std AIC=3.26
ged AIC=3.29
norm AIC=3.5
empirical
theoretical
0
5
10
−4
0
2
4
6
8
−2
−5
10
−3
−1
1
3



gjrGARCH(1,1)−jsu | VIX_chang
ret
t
Std residuals ._t
t
._t vs N(0,1)
Density
ACF . (WLB p=0.002)
ACF
Lag
ACF .² (WLB p=0.978)
ACF
Lag
Variance ratio=0.97 dev=−2.5%
cond s^2
emp var
model uncond
s^2
t

tep 8 — PIT density fits to ._t — VIX_




NO KS p=0
TF KS p=0
LO KS p=0
SST KS p=0.135
ST3 KS p=0.135
* JSU KS p=0.333
empirical
fitted
0
5
10
−3
−1
1
3
−10
−5
0
5
10
−4
−2
0
2
4
−5
0
5
10
−4
0
4
8